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Huang et al.

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(54) **CRYSTALLINE (R)-5-CARBAMOYLPIRIDIN-3-YL-2-METHYL-4-(3-(TRIFLUOROMETHOXY)BENZYL)PIPERAZINE-1-CARBOXYLATE, COMPOSITIONS AND METHODS OF USE THEREOF**

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Related U.S. Application Data

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(60) Provisional application No. 62/731,014, filed on Sep. 13, 2018.

(51) **Int. Cl.**
C07D 211/94 (2006.01)
A61P 35/00 (2006.01)

(52) **U.S. Cl.**
CPC **C07D 211/94** (2013.01); **A61P 35/00** (2018.01); **C07B 2200/13** (2013.01)

(58) **Field of Classification Search**
CPC C07D 211/94; A61P 35/00
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,053,199 B2 *	7/2021	Huang		A61P 35/00
11,680,046 B2	6/2023	Huang et al.		
2010/0009971 A1	1/2010	Ishii et al.		
2018/0256566 A1	9/2018	Grice et al.		
2019/0282568 A1	9/2019	Grice et al.		
2020/0095201 A1	3/2020	Huang et al.		
2021/0363105 A1	11/2021	Huang et al.		

FOREIGN PATENT DOCUMENTS

WO WO 2018/169880 * 9/2018 A61K 31/497

OTHER PUBLICATIONS

Bernstein, 2004, "Crystal Structure Prediction and Polymorphism," ACA Transactions, 39:14-23.

Braga et al., 2005, "Making crystals from crystals: a green route to crystal engineering and polymorphism," Chemical Communications, 3635-3645.

Chemburkar et al., 2000, "Dealing with the Impact of Ritonavir Polymorphism on the Late Stages of Bulk Drug Process Development," Organic Process Research & Development, 4(5):413-417.

Di Martino et al., 1997, "Preparation and Physical Characterization of Forms II and III of Paracetamol," Journal of Thermal Analysis, 48(3):447-458.

International Search Report and Written Opinion dated Nov. 14, 2019 for PCT/US2019/050771 (9 pages).

Jones et al., 2011, "Pharmaceutical Cocrystals: An Emerging Approach to Physical Property Enhancement," MRS Bulletin, 31:875-879.

Knapman et al., 2000, "Polymorphic predictions," Modern Drug Discovery, 3:53-57.

Price, 2004, "The computational prediction of pharmaceutical crystal structures and polymorphism," Advanced Drug Delivery Reviews, 56(3):301-319.

* cited by examiner

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(57) **ABSTRACT**

Provided herein are crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Pharmaceutical compositions comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are also disclosed.

16 Claims, 14 Drawing Sheets

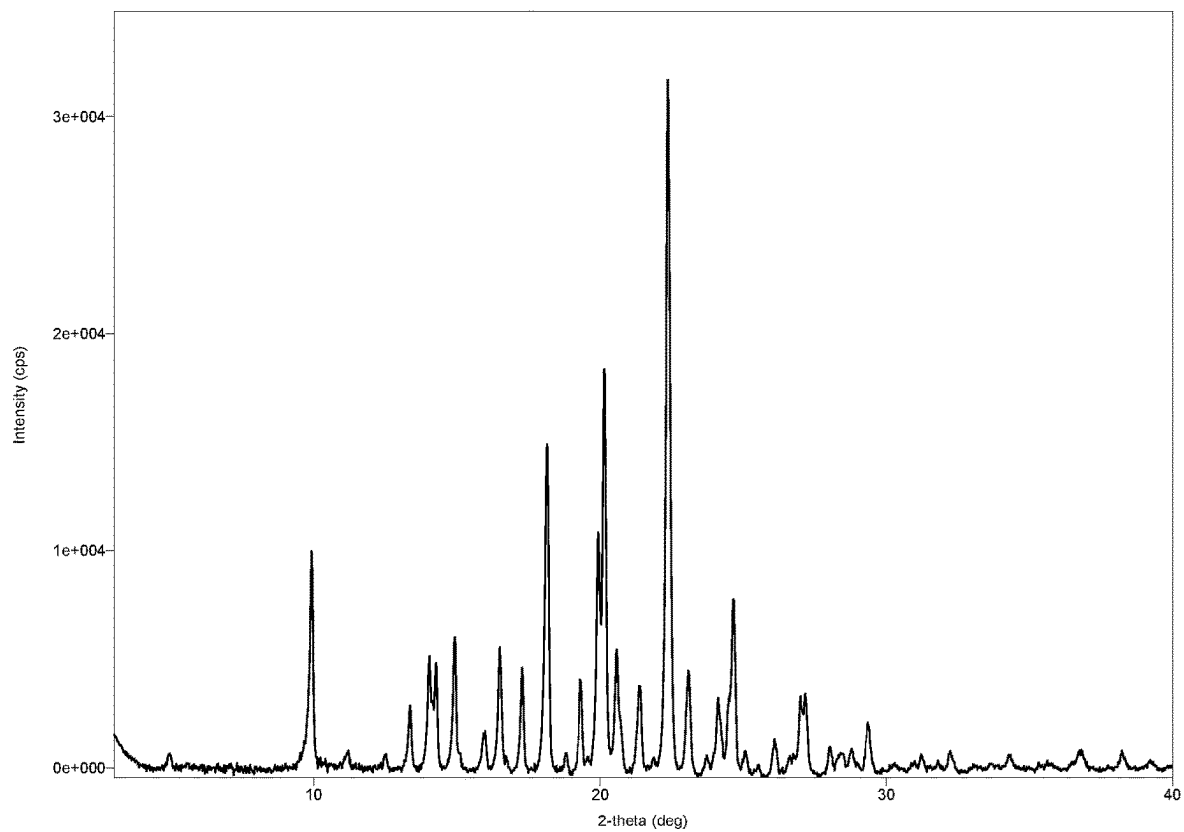
FIGURE 1

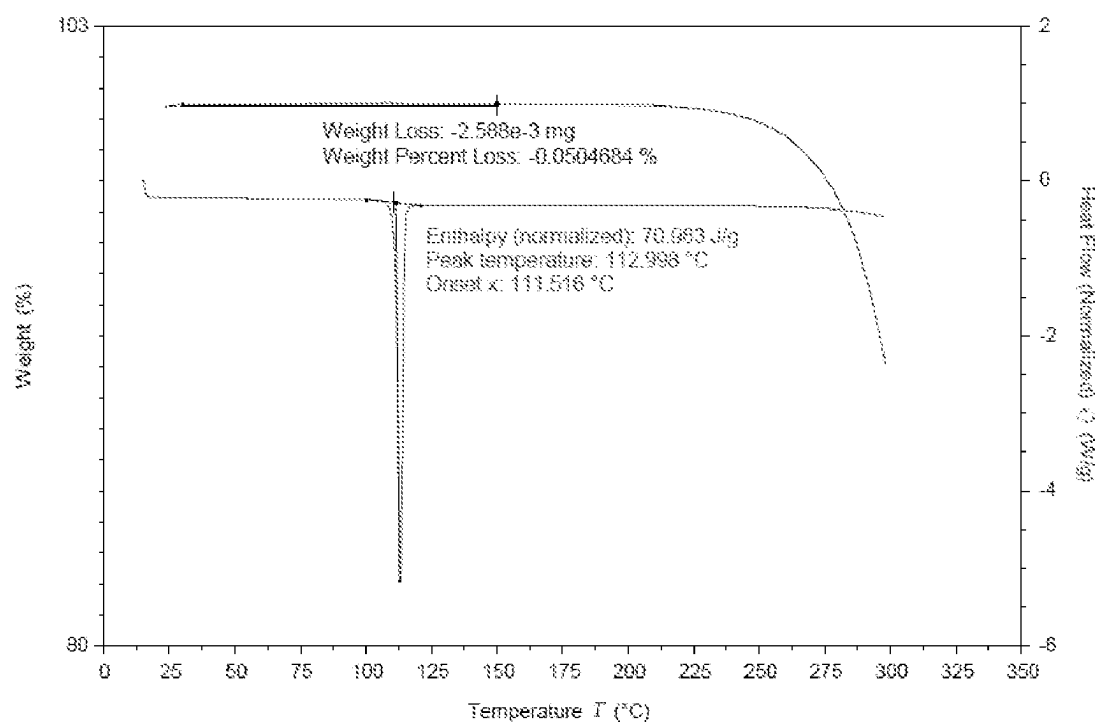
FIGURE 2

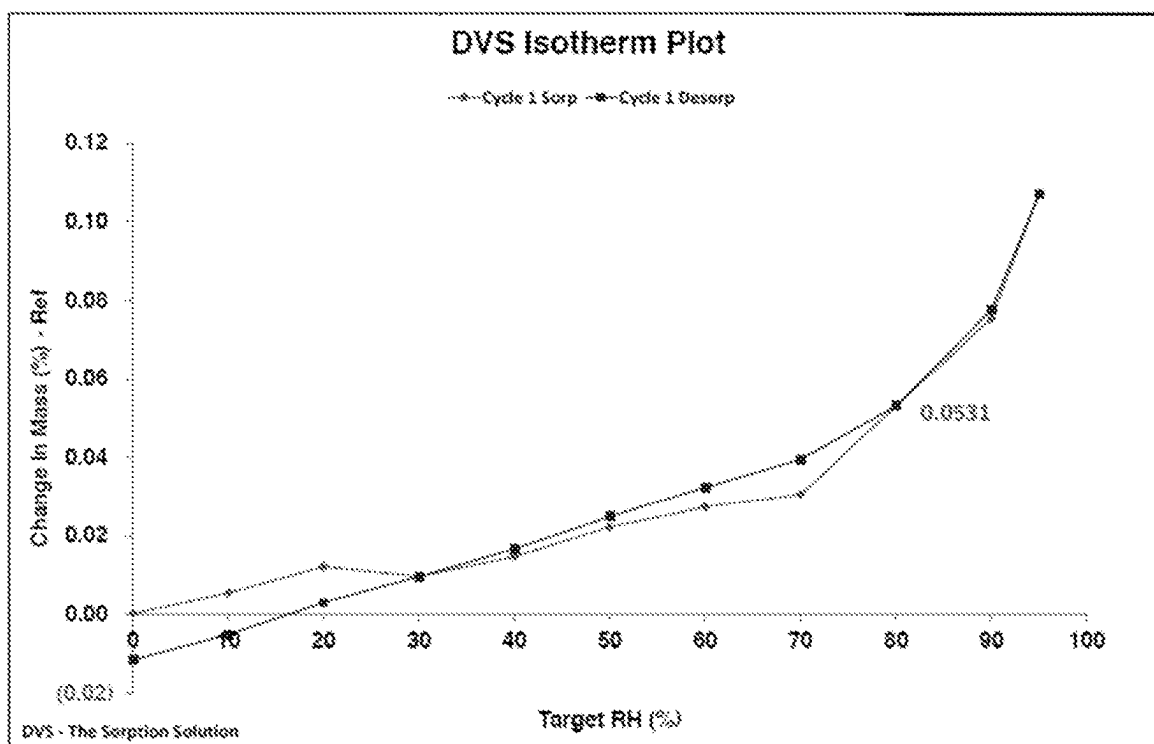
FIGURE 3

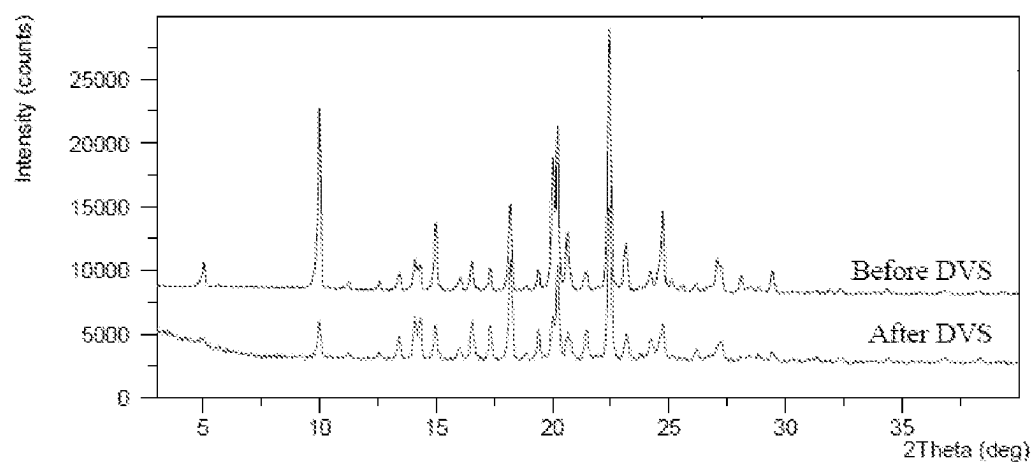
FIGURE 4

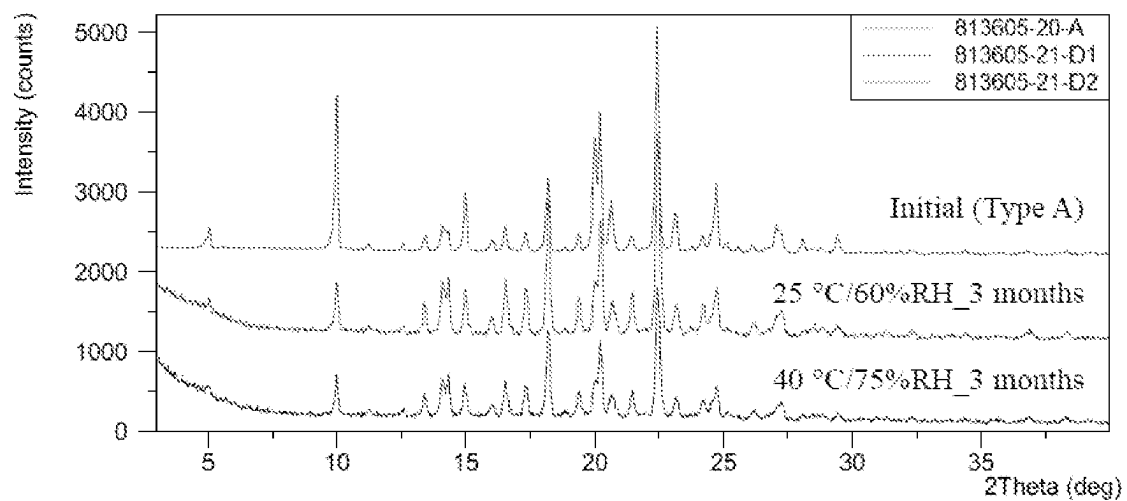
FIGURE 5

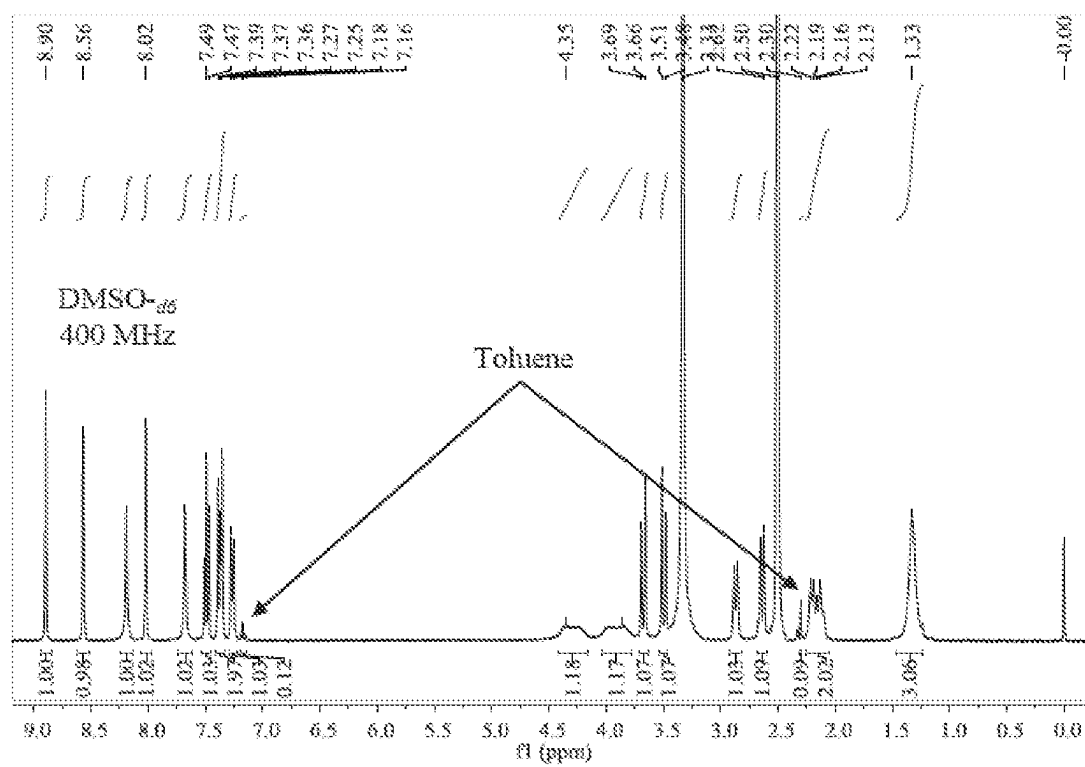
FIGURE 6

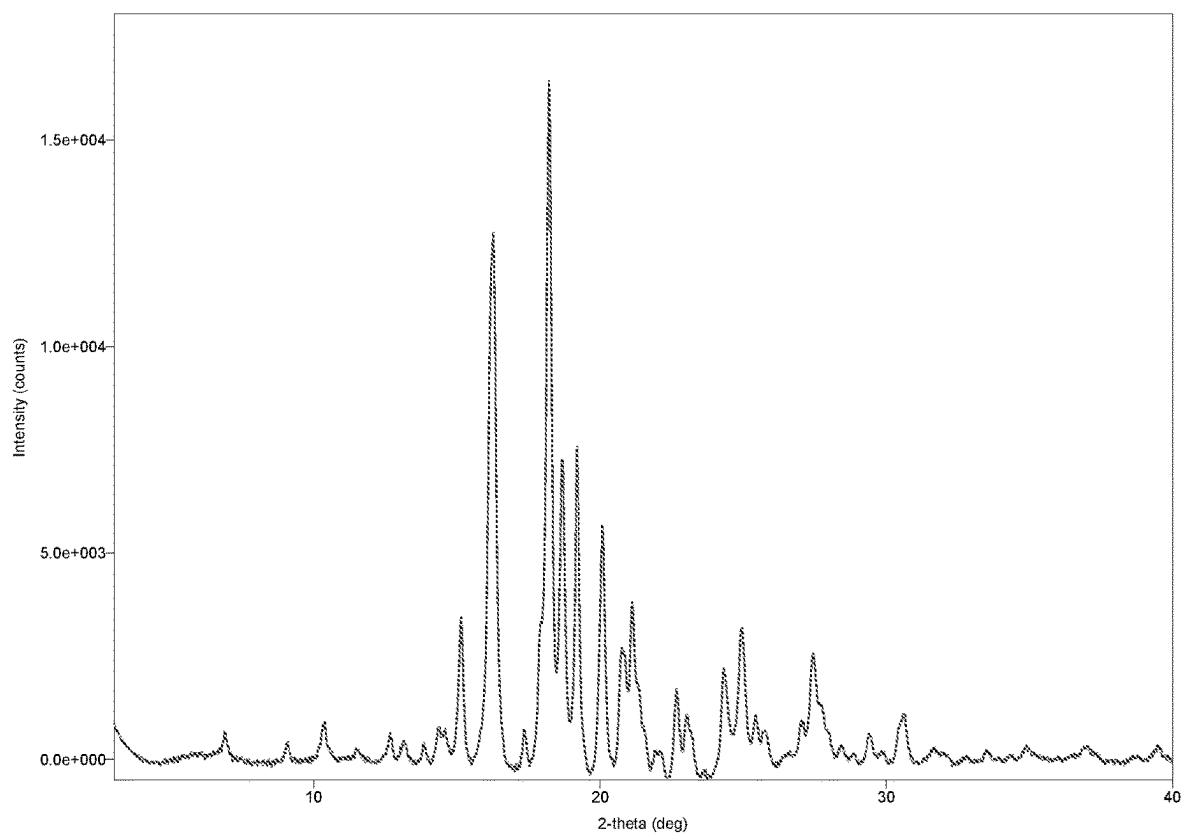
FIGURE 7

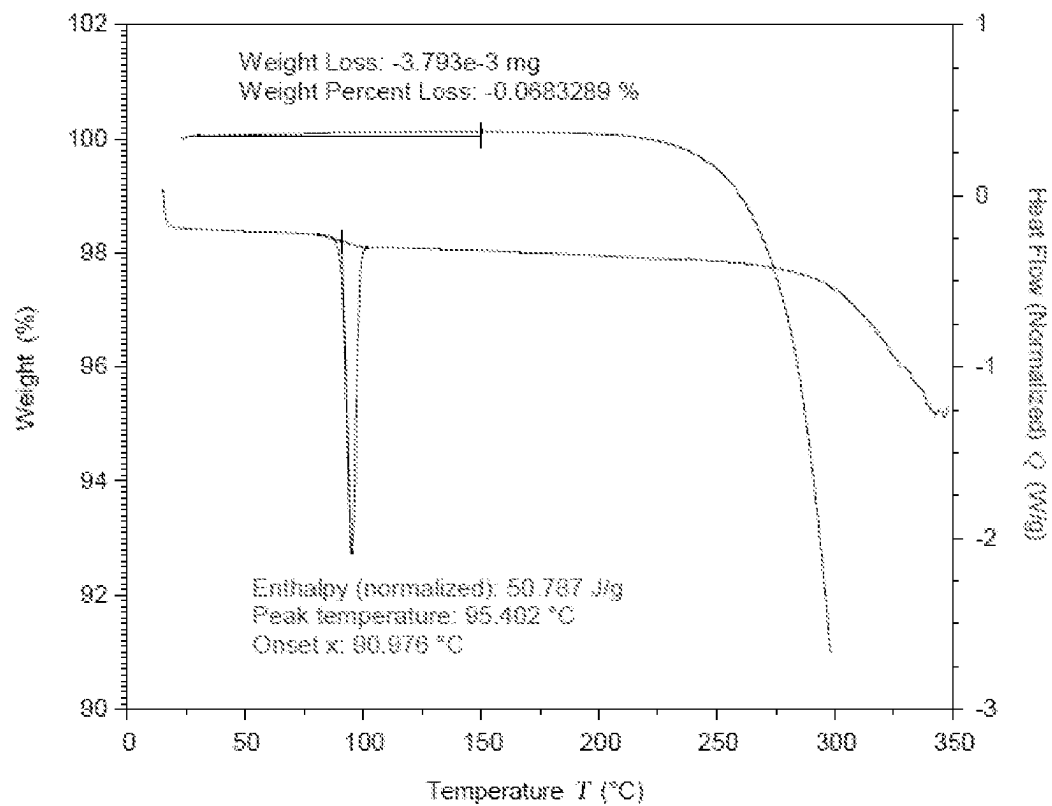
FIGURE 8

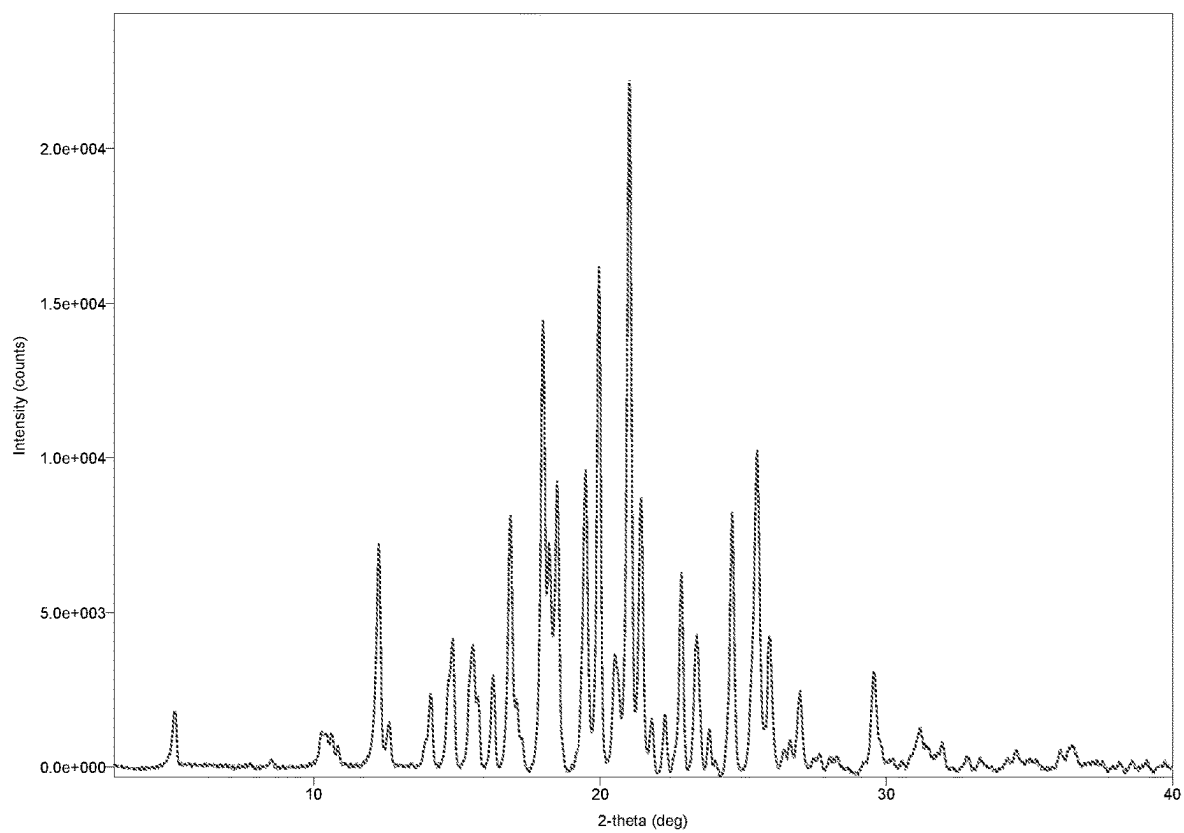
FIGURE 9

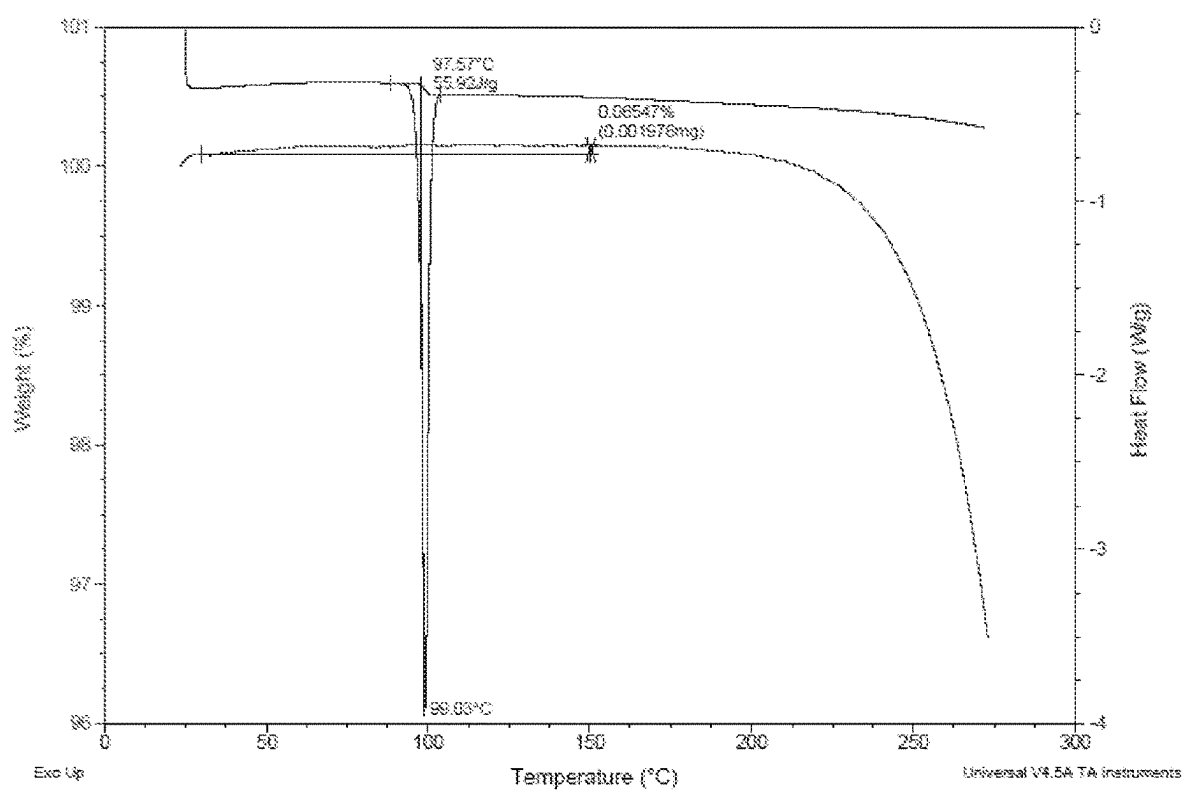
FIGURE 10

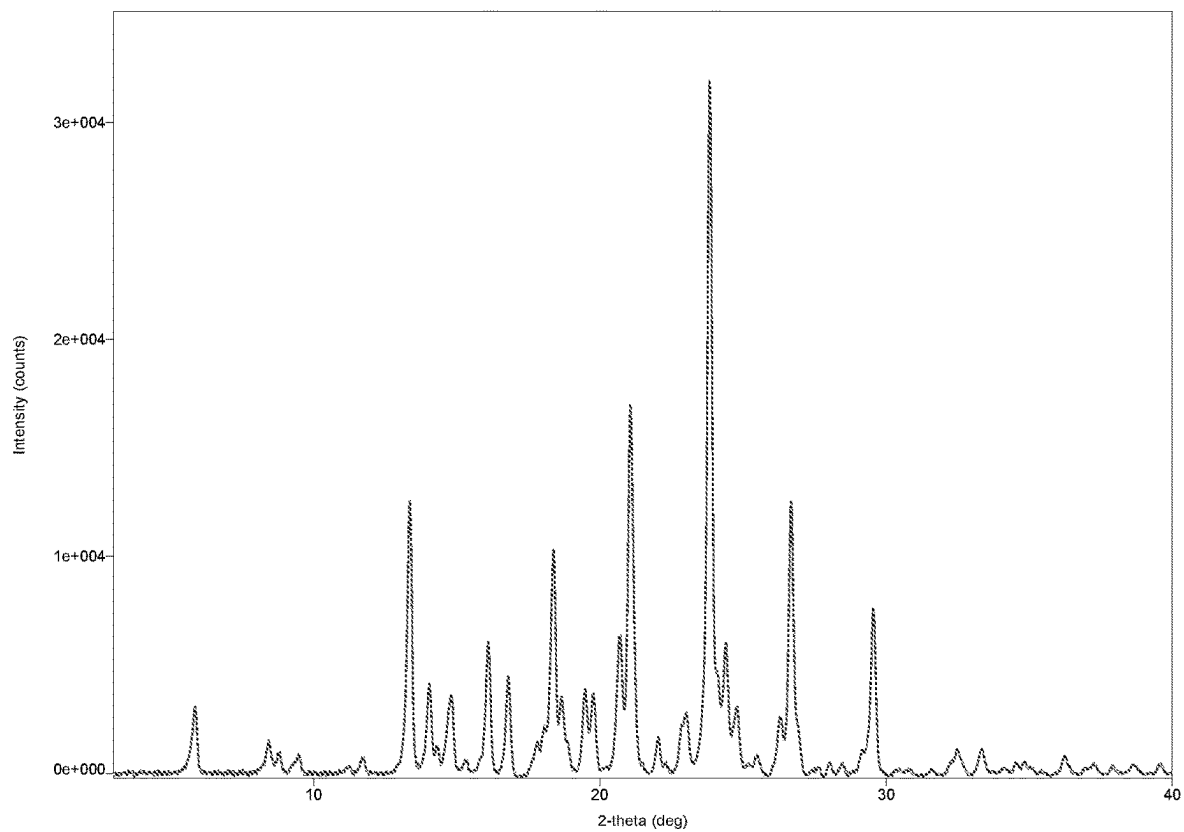
FIGURE 11

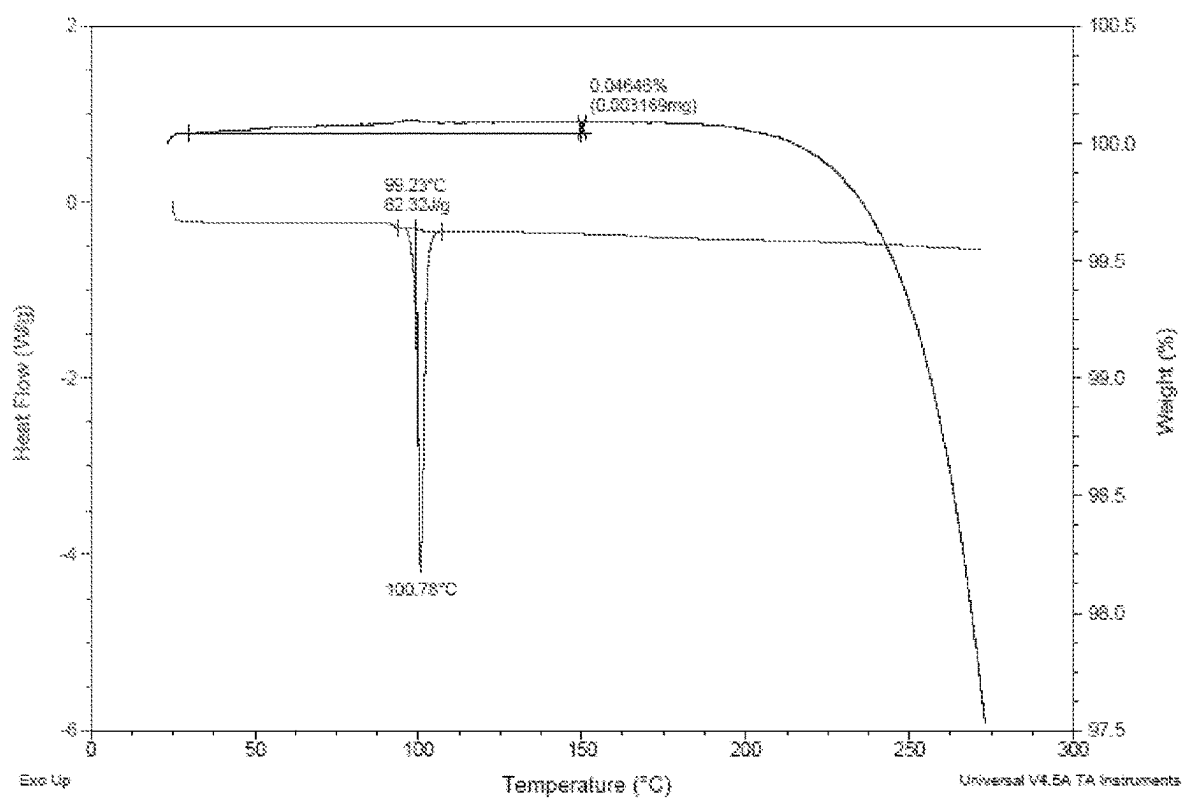
FIGURE 12

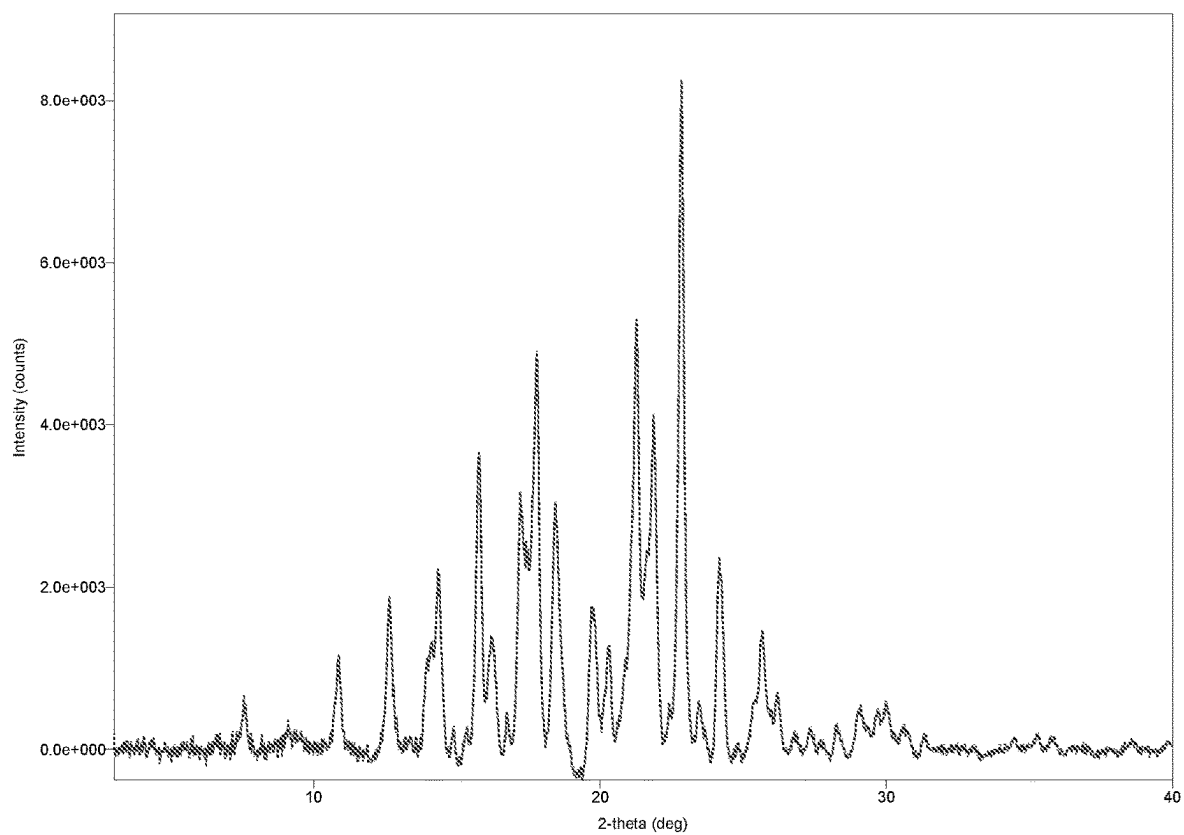
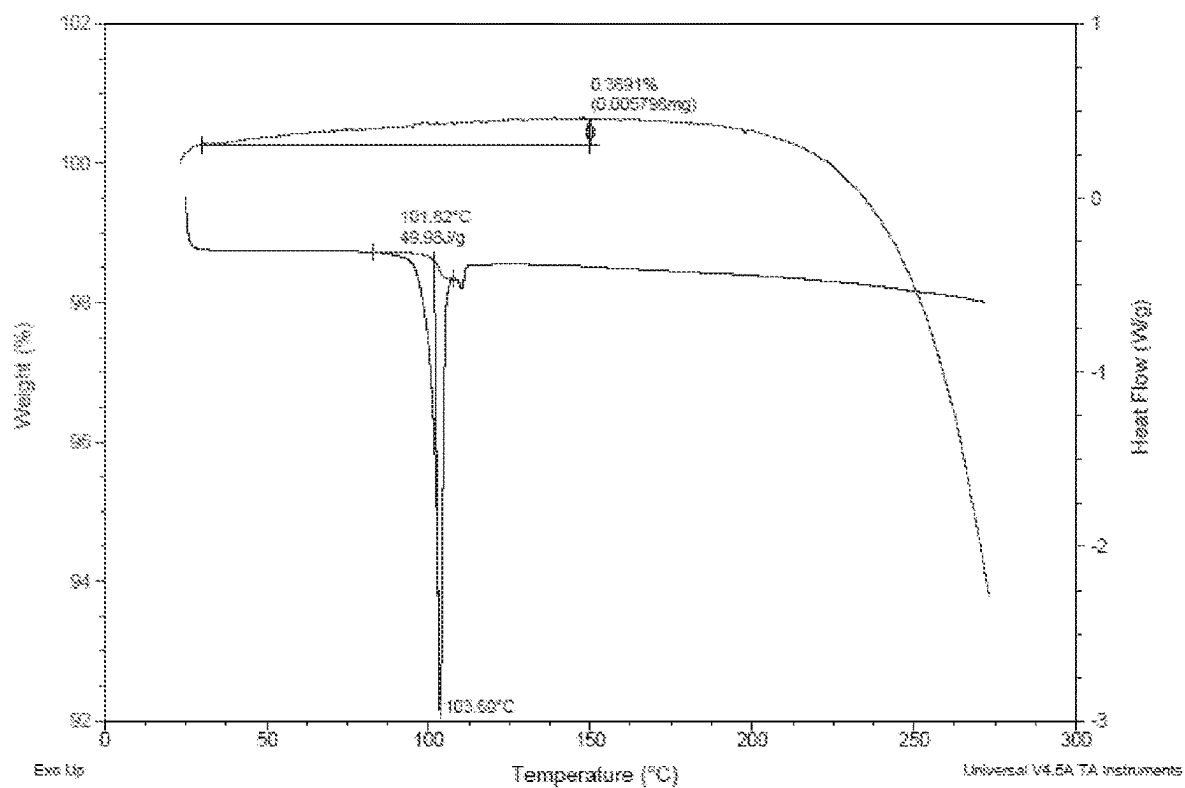
FIGURE 13

FIGURE 14

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**CRYSTALLINE
(R)-5-CARBAMOYLPIRIDIN-3-
YL-2-METHYL-4-(3-(TRIFLUOROMETHOXY)
BENZYL)PIPERAZINE-1-CARBOXYLATE,
COMPOSITIONS AND METHODS OF USE
THEREOF**

CROSS-REFERENCE

This application is a continuation of U.S. application Ser. No. 17/338,182, filed Jun. 3, 2021, which is a division of U.S. application Ser. No. 16/569,578, filed Sep. 12, 2019, now U.S. Pat. No. 11,053,199, which claims priority to U.S. Provisional Application No. 62/731,014, filed Sep. 13, 2018, each of which is herein incorporated by reference in its entirety.

FIELD

Provided herein are solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Pharmaceutical compositions comprising such solid forms and methods of use for treating, preventing, and managing various disorders are also provided herein.

BACKGROUND

Many compounds can exist in different crystal forms, or polymorphs, which exhibit different physical, chemical, and spectroscopic properties. For example, certain polymorphs of a compound may be more readily soluble in particular solvents, may flow more readily, or may compress more easily than others. See, e.g., P. DiMartino, et al., *J. Thermal Anal.*, 48:447-458 (1997). In the case of drugs, certain solid forms may be more bioavailable than others, while others may be more stable under certain manufacturing, storage, and biological conditions.

Polymorphic forms of a compound are known in the pharmaceutical arts to affect, for example, the solubility, stability, flowability, fractability, and compressibility of the compound, as well as the safety and efficacy of drug products comprising it. See, e.g., Knapman, K. *Modern Drug Discoveries*, 2000, 53. Therefore, the discovery of new polymorphs of a drug can provide a variety of advantages.

The identification and selection of a solid form of a pharmaceutical compound are complex, given that a change in solid form may affect a variety of physical and chemical properties, which may provide benefits or drawbacks in processing, formulation, stability, bioavailability, storage, handling (e.g., shipping), among other important pharmaceutical characteristics. Useful pharmaceutical solids include crystalline solids and amorphous solids, depending on the product and its mode of administration. Amorphous solids are characterized by a lack of long-range structural order, whereas crystalline solids are characterized by structural periodicity. The desired class of pharmaceutical solid depends upon the specific application; amorphous solids are sometimes selected on the basis of, e.g., an enhanced dissolution profile, while crystalline solids may be desirable for properties such as, e.g., physical or chemical stability.

The importance of discovering polymorphs was underscored by the case of Ritonavir™, an HIV protease inhibitor that was formulated as soft gelatin capsules. About two years after the product was launched, the unanticipated precipitation of a new, less soluble polymorph in the formulation necessitated the withdrawal of the product from the market

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until a more consistent formulation could be developed (see S. R. Chemburkar et al., *Org. Process Res. Dev.*, (2000) 4:413-417).

Notably, it is not possible to predict a priori if crystalline forms of a compound even exist, let alone how to successfully prepare them (see, e.g., Braga and Grepioni, 2005, "Making crystals from crystals: a green route to crystal engineering and polymorphism," *Chem. Commun.* :3635-3645 (with respect to crystal engineering, if instructions are not very precise and/or if other external factors affect the process, the result can be unpredictable); Jones et al., 2006, *Pharmaceutical Cocrystals: An Emerging Approach to Physical Property Enhancement*, "MRS Bulletin" 31:875-879 (At present it is not generally possible to computationally predict the number of observable polymorphs of even the simplest molecules); Price, 2004, "The computational prediction of pharmaceutical crystal structures and polymorphism," *Advanced Drug Delivery Reviews* 56:301-319 ("Price"); and Bernstein, 2004, "Crystal Structure Prediction and Polymorphism," *ACA Transactions* 39:14-23 (a great deal still needs to be learned and done before one can state with any degree of confidence the ability to predict a crystal structure, much less polymorphic forms)).

The variety of possible solid forms creates potential diversity in physical and chemical properties for a given pharmaceutical compound. The discovery and selection of solid forms are of great importance in the development of an effective, stable and marketable pharmaceutical product.

(R)-5-Carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate is a compound which is a dual MAGL and FAAH inhibitor, and may be used in treating various disorders.

The synthesis of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate has been described in PCT/US2018/022049 filed Mar. 12, 2018.

Novel crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are described herein. New polymorphic forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can further the development of formulations for the treatment of various illnesses, and may yield numerous formulation, manufacturing and therapeutic benefits.

SUMMARY

Provided herein are crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Also provided herein are pharmaceutical compositions comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Further provided herein are methods of treating or preventing a variety of disease and disorders, which comprise administering to a patient a therapeutically effective amount of a crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

Also provided herein are methods of preparing, isolating, and characterizing crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 provides a representative X-ray powder diffraction (XRPD) pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

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FIG. 2 provides a representative differential scanning calorimetry (DSC) thermogram and a representative thermogravimetric analysis (TGA) thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

FIG. 3 provides a representative dynamic vapor sorption (DVS) plot of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

FIG. 4 provides representative XRPD patterns of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A before and after a DVS experiment.

FIG. 5 provides representative XRPD patterns of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A before and after accelerated conditions stability tests.

FIG. 6 provides a representative nuclear magnetic resonance (NMR) spectrum of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

FIG. 7 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

FIG. 8 provides a representative DSC thermogram and a representative TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

FIG. 9 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

FIG. 10 provides a representative DSC thermogram and TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

FIG. 11 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

FIG. 12 provides a representative DSC thermogram and TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

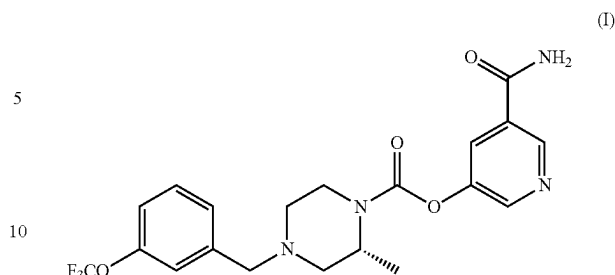
FIG. 13 provides a representative XRPD pattern of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

FIG. 14 provides a representative DSC thermogram and TGA thermogram of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

DEFINITIONS

As used herein, and unless otherwise specified, the compound referred to herein as (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate corresponds to a compound of Formula (I), depicted below.

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(R)-5-Carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained via standard, synthetic methods (see e.g., PCT/US2018/022049).

Unless otherwise specified, the term “crystalline” and related terms used herein, when used to describe a substance, component, product, or form, mean that the substance, component, product, or form is substantially crystalline, for example, as determined by X-ray diffraction. (see, e.g., *Remington's Pharmaceutical Sciences*, 20th ed., Lippincott Williams & Wilkins, Philadelphia PA, 173 (2000); *The United States Pharmacopeia*, 37th ed., 503-509 (2014)).

As used herein, and unless otherwise specified, the terms “about” and “approximately,” when used in connection with doses, amounts, or weight percents of ingredients of a composition or a dosage form, mean a dose, amount, or weight percent that is recognized by one of ordinary skill in the art to provide a pharmacological effect equivalent to that obtained from the specified dose, amount, or weight percent. In certain embodiments, the terms “about” and “approximately,” when used in this context, contemplate a dose, amount, or weight percent within 30%, within 20%, within 15%, within 10%, or within 5%, of the specified dose, amount, or weight percent.

As used herein, and unless otherwise specified, the terms “about” and “approximately,” when used in connection with a numeric value or range of values which is provided to characterize a particular solid form, e.g., a specific temperature or temperature range, such as, for example, that describes a melting, dehydration, desolvation, or glass transition temperature; a mass change, such as, for example, a mass change as a function of temperature or humidity; a solvent or water content, in terms of, for example, mass or a percentage; or a peak position, such as, for example, in analysis by, for example, IR or Raman spectroscopy or XRPD; indicate that the value or range of values may deviate to an extent deemed reasonable to one of ordinary skill in the art while still describing the solid form. Techniques for characterizing crystal forms and amorphous forms include, but are not limited to, thermal gravimetric analysis (TGA), differential scanning calorimetry (DSC), X-ray powder diffractometry (XRPD), single-crystal X-ray diffractometry, vibrational spectroscopy, e.g., infrared (IR) and Raman spectroscopy, solid-state and solution nuclear magnetic resonance (NMR) spectroscopy, optical microscopy, hot stage optical microscopy, scanning electron microscopy (SEM), electron crystallography and quantitative analysis, particle size analysis (PSA), surface area analysis, solubility studies, and dissolution studies. In certain embodiments, the terms “about” and “approximately,” when used in this context, indicate that the numeric value or range of values may vary within 30%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1.5%, 1%, 0.5%, or of the

recited value or range of values. In the context of molar ratios, “about” and “approximately” indicate that the numeric value or range of values may vary within 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1.5%, 1%, 0.5%, or 0.25% of the recited value or range of values. It should be understood that the numerical values of the peaks of an X-ray powder diffraction pattern may vary from one machine to another, or from one sample to another, and so the values quoted are not to be construed as absolute, but with an allowable variability, such as ± 0.2 degrees two theta ($^{\circ} 2\theta$), or more. For example, in some embodiments, the value of an XRPD peak position may vary by up to ± 0.2 degrees 2θ while still describing the particular XRPD peak.

As used herein, and unless otherwise specified, a solid form that is “substantially physically pure” is substantially free from other chemical forms. In certain embodiments, a crystal form that is substantially physically pure contains less than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, or 0.01% of one or more other solid forms on a weight basis. The detection of other solid forms can be accomplished by any method apparent to a person of ordinary skill in the art, including, but not limited to, diffraction analysis, thermal analysis, elemental combustion analysis and/or spectroscopic analysis.

As used herein, and unless otherwise specified, a solid form that is “substantially chemically pure” is substantially free from other chemical compounds (i.e., chemical impurities). In certain embodiments, a solid form that is substantially chemically pure contains less than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.01% of one or more other chemical compounds on a weight basis. The detection of other chemical compounds can be accomplished by any method apparent to a person of ordinary skill in the art, including, but not limited to, methods of chemical analysis, such as, e.g., mass spectrometry analysis, spectroscopic analysis, thermal analysis, elemental combustion analysis and/or chromatographic analysis.

As used herein, and unless otherwise indicated, a chemical compound, solid form, or composition that is “substantially free” of another chemical compound, solid form, or composition means that the compound, solid form, or composition contains, in certain embodiments, less than about 50%, 45%, 40%, 35%, 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.4%, 0.3%, 0.2%, 0.1%, 0.05%, or 0.01% by weight of the other compound, solid form, or composition.

Unless otherwise specified, the terms “anhydrate” and “anhydrous” as used herein, refer to a solid form of a substance which does not contain water in its crystal lattice.

As used herein, and unless otherwise specified, the terms “treat,” “treating” and “treatment” refer to the eradication or amelioration of a disease or disorder, or of one or more symptoms associated with the disease or disorder. In certain embodiments, the terms refer to minimizing the spread or worsening of the disease or disorder resulting from the administration of one or more prophylactic or therapeutic agents to a subject with such a disease or disorder. In some embodiments, the terms refer to the administration of a compound provided herein, with or without other additional active agents, after the onset of symptoms of a particular disease.

Unless otherwise specified, the term “composition” as used herein is intended to encompass a product comprising the specified ingredient(s) (and in the specified amount(s), if

indicated), as well as any product which results, directly or indirectly, from combination of the specified ingredient(s) in the specified amount(s). By “pharmaceutically acceptable,” it is meant a diluent, excipient, or carrier in a formulation must be compatible with the other ingredient(s) of the formulation and not deleterious to the recipient thereof.

Unless otherwise specified, the term “subject” is defined herein to include animals, such as mammals, including, but not limited to, primates (e.g., humans), cows, sheep, goats, horses, dogs, cats, rabbits, rats, mice, and the like. In specific embodiments, the subject is a human.

Unless otherwise specified, to the extent that there is a discrepancy between a depicted chemical structure of a compound provided herein and a chemical name of a compound provided herein, the chemical structure shall control.

DETAILED DESCRIPTION

Provided herein are solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. In certain embodiments, the solid form is crystalline. In certain embodiments, the solid form is a single-component solid form. In certain embodiments, the solid form is an anhydrate.

Crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be prepared by the methods described herein, including the methods described in the Examples below, or by techniques known in the art, including heating, cooling, freeze drying, lyophilization, solvent evaporation, solvent recrystallization, antisolvent addition, slurry recrystallization, rapid cooling, slow cooling, exposure to solvent and/or water, and drying. The particle size of the resulting solid forms, which can vary, e.g., from nanometer dimensions to millimeter dimensions, can be controlled, e.g., by varying crystallization conditions, such as, e.g., the rate of crystallization and/or the crystallization solvent system, or by particle-size reduction techniques, e.g., grinding, milling, micronizing or sonication.

While not intending to be bound by any particular theory, crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are characterized by physical properties, e.g., stability, solubility and dissolution rate, appropriate for pharmaceutical and therapeutic dosage forms. Moreover, while not wishing to be bound by any particular theory, crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are characterized by physical properties (e.g., density, compressibility, hardness, morphology, cleavage, stickiness, solubility, water uptake, electrical properties, thermal behavior, solid-state reactivity, physical stability, and chemical stability) affecting particular processes (e.g., yield, filtration, washing, drying, milling, mixing, tableting, flowability, dissolution, formulation, and lyophilization) which make certain crystalline forms suitable for the manufacture of a dosage form. Such properties can be determined using particular analytical chemical techniques, including solid-state analytical techniques (e.g., X-ray diffraction, microscopy, spectroscopy and thermal analysis), as described herein and known in the art.

Certain embodiments herein provide compositions comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

Certain embodiments provide methods of using these compositions in the treatment, prevention or management of

diseases and disorders including, but not limited to, diseases and disorders related to MAGL and/or FAAH activity.

Certain embodiments herein provide methods of using crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate in the treatment, prevention, or management of pain, neurological disorders, anxiety, inflammatory bowel disease, neuropathic pain, and the proliferation and migration of cancer cells.

Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from methanol. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from isopropyl acetate. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from a 9:1 ethanol/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A can be obtained from toluene.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A may be characterized by X-ray powder diffraction analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern comprising peaks at 16.5, 20.1, and 22.3 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern comprising peaks at 16.4, 20.1, and 22.3 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern comprising peaks at 18.1, 20.1, and 22.3 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A wherein the X-ray powder diffraction pattern further comprises peaks at 9.9, 19.9, and 24.6 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A wherein the X-ray powder diffraction pattern further comprises peaks at 9.9, 19.9, and 24.7 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, or fourteen of the following approximate positions: 9.9, 13.3, 13.9, 14.8, 16.4, 18.0, 19.9, 20.1, 22.3, 23.0, 24.4, 24.6, 26.9, and 29.3 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17,

18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen, nineteen, twenty, or twenty-one of the following approximate positions: 9.9, 13.4, 14.0, 14.2, 14.9, 16.0, 16.5, 17.3, 18.1, 19.3, 19.9, 20.1, 20.6, 21.4, 22.3, 23.1, 24.1, 24.7, 27.0, 27.2, and 29.3 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction pattern depicted in FIG. 1.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A may be characterized by thermal analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 113° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 112° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 112° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 111° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermogram depicted in FIG. 2.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A which is anhydrous.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.22% when heated from about 25° C. to about 100° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 2.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A which is substantially physically pure.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A which is substantially chemically pure.

In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A.

Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B can be obtained from a 9:1 acetonitrile/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B can be obtained from an ethyl acetate/n-heptane mixture, methyl acetate, acetone or methylene chloride.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B may be characterized by X-ray powder diffraction analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having an X-ray powder diffraction pattern comprising peaks at 16.2, 18.2, and 19.2 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B wherein the X-ray powder diffraction pattern further comprises peaks at 15.1, 18.7, and 20.1 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve of the following approximate positions: 15.1, 16.2, 17.9, 18.2, 18.7, 19.2, 20.1, 21.1, 24.3, 25.0, 27.4, and 27.6 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 7.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B may be characterized by thermal analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 95° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 91° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermogram depicted in FIG. 8.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B which is anhydrous.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram comprising a weight loss of less than about 2.5% when heated from about 30° C. to about 100° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 8.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B which is substantially physically pure.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B which is substantially chemically pure.

In one embodiment, provided is a pharmaceutical composition comprising a crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B.

Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C can be obtained from a 1:3 chloroform/hexane mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-car-

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boxylate can be obtained from a 1:1 ethanol/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from a 3:1 acetonitrile/water mixture. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from slurrying Form B in FaSSiF. In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate can be obtained from slurrying Form B in FeSSiF.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C may be characterized by X-ray powder diffraction analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having an X-ray powder diffraction pattern comprising peaks at 18.0, 20.0, and 21.0 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C wherein the X-ray powder diffraction pattern further comprises peaks at 12.3, 16.8, and 25.5 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen, nineteen, twenty, twenty-one, or twenty-two of the following approximate positions: 12.3, 14.1, 14.7, 14.8, 15.5, 15.7, 16.2, 16.8, 17.1, 18.0, 18.2, 18.5, 19.5, 20.0, 20.5, 21.0, 21.4, 22.8, 24.6, 25.5, 25.9, and 29.6 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 9.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C may be characterized by thermal analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 99° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 98° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermograms depicted in FIG. 10.

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In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C which is anhydrous.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram comprising a weight loss of about 0.149% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram comprising a weight loss of about 0.15% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 10.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C which is substantially physically pure.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C which is substantially chemically pure.

In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C.

Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D can be obtained from slurrying Form B in methyl tert-butyl ether (MTBE).

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D may be characterized by X-ray powder diffraction analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having an X-ray powder diffraction pattern comprising peaks at 13.4, 21.0, and 23.8 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D wherein the X-ray powder diffraction pattern further comprises peaks at 18.3, 26.7, and 29.5 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, or seventeen of the following approximate positions: 13.4, 14.0, 14.8, 16.1, 16.8, 18.0, 18.3, 18.6, 19.5, 19.8, 20.7, 21.0, 22.8, 23.8, 24.4, 26.7, and 29.5 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-

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methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 11.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D may be characterized by thermal analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 101° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 99° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermogram depicted in FIG. 12.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D which is anhydrous.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a thermogravimetric analysis thermogram comprising a weight loss of about 0.054% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 12.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D which is substantially physically pure.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D which is substantially chemically pure.

In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D.

Certain embodiments herein provide crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

In one embodiment provided herein, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)

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benzyl)piperazine-1-carboxylate Form E can be obtained from a chloroform/hexane mixture.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E may be characterized by X-ray powder diffraction analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having an X-ray powder diffraction pattern comprising peaks at 17.8, 21.3, and 22.8 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E wherein the X-ray powder diffraction pattern further comprises peaks at 15.8, 18.4, and 21.9 degrees 2 θ $\pm$ 0.2 degrees 2 θ .

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E is characterized by XRPD peaks located at one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve of the following approximate positions: 10.9, 12.7, 15.8, 17.2, 17.8, 18.4, 19.7, 21.3, 21.9, 22.8, 24.2, and 25.6 degrees 2 θ . In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate is characterized by an XRPD pattern having 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 peaks matching peaks in the representative XRPD pattern provided herein.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having an X-ray powder diffraction pattern corresponding to the representative X-ray powder diffraction patterns depicted in FIG. 13.

In certain embodiments, crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E may be characterized by thermal analysis.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a differential scanning calorimetry thermogram comprising an endotherm with a maximum at about 104° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a differential scanning calorimetry thermogram comprising an endotherm with an onset at about 102° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a differential scanning calorimetry thermogram corresponding to the representative differential scanning calorimetry thermograms depicted in FIG. 14.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E which is anhydrous.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a thermogravimetric analysis thermogram comprising a weight loss of about 0.37% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a thermogravimetric

ric analysis thermogram comprising a weight loss of less than about 1.0% when heated from about 30° C. to about 150° C.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E having a thermogravimetric analysis thermogram corresponding to the representative thermogravimetric analysis thermogram depicted in FIG. 14.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E which is substantially physically pure.

In one embodiment, provided is crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E which is substantially chemically pure.

In one embodiment, provided is a pharmaceutical composition comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E.

Pharmaceutical Compositions

Pharmaceutical compositions and single unit dosage forms comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate are provided herein. Also provided herein are methods for preparing pharmaceutical compositions and single unit dosage forms comprising crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate.

For example, in certain embodiments, individual dosage forms comprising crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate provided herein or prepared using crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate provided herein may be suitable for oral, mucosal (including rectal, nasal, or vaginal), parenteral (including subcutaneous, intramuscular, bolus injection, intraarterial, or intravenous), sublingual, transdermal, buccal, or topical administration.

In certain embodiments, pharmaceutical compositions and dosage forms provided herein comprise crystalline forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Certain embodiments herein provide pharmaceutical compositions and dosage forms comprising crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, wherein the crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate is substantially pure. Certain embodiments herein provide pharmaceutical compositions and dosage forms comprising crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate and/or amorphous solid forms of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate. Pharmaceutical compositions and dosage forms provided herein typically also comprise one or more pharmaceutically acceptable excipients, diluents or carriers.

Single unit dosage forms provided herein are suitable for oral or parenteral (e.g., subcutaneous, intravenous, bolus injection, intramuscular, or intraarterial) administration to a patient. Examples of dosage forms include, but are not limited to: tablets; caplets; capsules, such as soft elastic gelatin capsules; powders and sterile solids that can be reconstituted to provide liquid dosage forms suitable for parenteral administration to a patient.

The composition, shape, and type of dosage forms provided herein will typically vary depending on their use. A parenteral dosage form may contain smaller amounts of one or more of the active ingredients it comprises than an oral dosage form used to treat the same disease or disorder. These and other ways in which specific dosage forms encompassed by this invention will vary from one another will be readily apparent to those skilled in the art. See, e.g., *Remington's Pharmaceutical Sciences*, 18th ed., Mack Publishing, Easton Pa. (1990).

Typical pharmaceutical compositions and dosage forms comprise one or more excipients. Suitable excipients are well known to those skilled in the art of pharmacy, and non-limiting examples of suitable excipients are provided herein. Whether a particular excipient is suitable for incorporation into a pharmaceutical composition or dosage form depends on a variety of factors well known in the art including, but not limited to, the way in which the dosage form will be administered to a patient. For example, oral dosage forms such as tablets may contain excipients not suited for use in parenteral dosage forms. The suitability of a particular excipient may also depend on the specific active ingredients in the dosage form.

Like the amounts and types of excipients, the amounts and specific types of active ingredients in a dosage form may differ depending on factors such as, but not limited to, the route by which it is to be administered to patients. However, typical dosage forms provided herein lie within the range of from about 0.1 mg to about 1,000 mg per day, given as a single once-a-day dose in the morning or as divided doses throughout the day. More specifically, the daily dose may be administered twice, three times, or four times daily in equally divided doses. Specifically, a daily dose range may be from about 0.1 mg to about 500 mg per day, more specifically, between about 0.1 mg and about 200 mg per day. A daily dose range may be 1 mg, 2 mg, 3 mg, 4 mg, or 5 mg. In managing the patient, the therapy may be initiated at a lower dose, perhaps about 1 mg to about 25 mg, and increased if necessary up to about 200 mg to about 1,000 mg per day as either a single dose or divided doses, depending on the patient's global response.

Oral Dosage Forms

Pharmaceutical compositions provided herein that are suitable for oral administration can be presented as discrete dosage forms, such as, but are not limited to, tablets (e.g., chewable tablets), caplets, capsules, and liquids (e.g., flavored syrups). Such dosage forms contain predetermined amounts of active ingredients, and may be prepared by methods of pharmacy well known to those skilled in the art. See generally *Remington's Pharmaceutical Sciences*, 18th ed., Mack Publishing, Easton Pa. (1990).

Typical oral dosage forms provided herein are prepared by combining the active ingredient(s) in an intimate admixture with at least one excipient according to conventional pharmaceutical compounding techniques. Excipients can take a wide variety of forms depending on the form of preparation desired for administration. For example, excipients suitable for use in oral liquid or aerosol dosage forms include, but are not limited to, water, glycols, oils, alcohols, flavoring agents, preservatives, and coloring agents. Examples of excipients suitable for use in solid oral dosage forms (e.g., powders, tablets, capsules, and caplets) include, but are not limited to, starches, sugars, micro-crystalline cellulose, diluents, granulating agents, lubricants, binders, and disintegrating agents.

If desired, tablets can be coated by standard aqueous or nonaqueous techniques. Such dosage forms can be prepared by any of the methods of pharmacy. In general, pharmaceutical compositions and dosage forms are prepared by uniformly and intimately admixing the active ingredients with

liquid carriers, finely divided solid carriers, or both, and then shaping the product into the desired presentation if necessary.

For example, a tablet can be prepared by compression or molding. Compressed tablets can be prepared by compressing in a suitable machine the active ingredients in a free-flowing form such as powder or granules, optionally mixed with an excipient. Molded tablets can be made by molding in a suitable machine a mixture of the powdered compound moistened with an inert liquid diluent.

Examples of excipients that can be used in oral dosage forms of the invention include, but are not limited to, binders, fillers, disintegrants, and lubricants. Binders suitable for use in pharmaceutical compositions and dosage forms include, but are not limited to, corn starch, potato starch, or other starches, gelatin, natural and synthetic gums such as acacia, sodium alginate, alginic acid, other alginates, powdered tragacanth, guar gum, cellulose and its derivatives (e.g., ethyl cellulose, cellulose acetate, carboxymethyl cellulose calcium, sodium carboxymethyl cellulose), polyvinyl pyrrolidone, methyl cellulose, pre-gelatinized starch, hydroxypropyl methyl cellulose, (e.g., Nos. 2208, 2906, 2910), microcrystalline cellulose, and mixtures thereof.

Examples of fillers suitable for use in the pharmaceutical compositions and dosage forms disclosed herein include, but are not limited to, talc, calcium carbonate (e.g., granules or powder), microcrystalline cellulose, powdered cellulose, dextrates, kaolin, mannitol, silicic acid, sorbitol, starch, pre-gelatinized starch, and mixtures thereof. The binder or filler in pharmaceutical compositions of the invention is typically present in from about 50 to about 99 weight percent of the pharmaceutical composition or dosage form.

Suitable forms of microcrystalline cellulose include, but are not limited to, the materials sold as AVICEL-PH-101™, AVICEL-PH-103™, AVICEL RC-581™, AVICEL-PH-105™ (available from FMC Corporation, American Viscose Division, Avicel Sales, Marcus Hook, Pa.), and mixtures thereof. A specific binder is a mixture of microcrystalline cellulose and sodium carboxymethyl cellulose sold as AVICEL RC-58™. Suitable anhydrous or low moisture excipients or additives include AVICEL-PH-103™ and Starch 1500 LM™.

Disintegrants are used in the compositions provided herein to provide tablets that disintegrate when exposed to an aqueous environment. Tablets that contain too much disintegrant may disintegrate in storage, while those that contain too little may not disintegrate at a desired rate or under the desired conditions. Thus, a sufficient amount of disintegrant that is neither too much nor too little to detrimentally alter the release of the active ingredients should be used to form solid oral dosage forms of the invention. The amount of disintegrant used varies based upon the type of formulation, and is readily discernible to those of ordinary skill in the art. Typical pharmaceutical compositions comprise from about 0.5 to about 15 weight percent of disintegrant, specifically from about 1 to about 5 weight percent of disintegrant.

Disintegrants that can be used in pharmaceutical compositions and dosage forms provided herein include, but are not limited to, agar-agar, alginic acid, calcium carbonate, microcrystalline cellulose, croscarmellose sodium, crospovidone, polacrillin potassium, sodium starch glycolate, potato or tapioca starch, pre-gelatinized starch, other starches, clays, other algin, other celluloses, gums, and mixtures thereof.

Lubricants that can be used in pharmaceutical compositions and dosage forms provided herein include, but are not limited to, calcium stearate, magnesium stearate, mineral oil, light mineral oil, glycerin, sorbitol, mannitol, polyethylene glycol, other glycols, stearic acid, sodium lauryl sulfate, talc, hydrogenated vegetable oil (e.g., peanut oil, cottonseed oil, sunflower oil, sesame oil, olive oil, corn oil, and soybean oil), zinc stearate, ethyl oleate, ethyl laureate, agar, and

mixtures thereof. Additional lubricants include, for example, a syloid silica gel (AEROSIL 200™, manufactured by W.R. Grace Co. of Baltimore, Md.), a coagulated aerosol of synthetic silica (marketed by Degussa Co. of Plano, Tex.), CAB-O-SIL™ (a pyrogenic silicon dioxide product sold by Cabot Co. of Boston, Mass.), and mixtures thereof. If used at all, lubricants are typically used in an amount of less than about one weight percent of the pharmaceutical compositions or dosage forms into which they are incorporated.

Parenteral Dosage Forms

Parenteral dosage forms can be administered to patients by various routes including, but not limited to, subcutaneous, intravenous (including bolus injection), intramuscular, and intraarterial injection. Because their administration typically bypasses patients' natural defenses against contaminants, parenteral dosage forms are preferably sterile or capable of being sterilized prior to administration to a patient. Examples of parenteral dosage forms include, but are not limited to, solutions ready for injection, dry products ready to be dissolved or suspended in a pharmaceutically acceptable vehicle for injection, suspensions ready for injection, and emulsions.

Suitable vehicles that can be used to provide parenteral dosage forms provided herein are well known to those skilled in the art. Examples include, but are not limited to: Water for Injection USP; aqueous vehicles such as, but not limited to, Sodium Chloride Injection, Ringer's Injection, Dextrose Injection, Dextrose and Sodium Chloride Injection, and Lactated Ringer's Injection; water-miscible vehicles such as, but not limited to, ethyl alcohol, polyethylene glycol, and polypropylene glycol; and non-aqueous vehicles such as, but not limited to, corn oil, cottonseed oil, peanut oil, sesame oil, ethyl oleate, isopropyl myristate, and benzyl benzoate.

Compounds that increase the solubility of one or more of the active ingredients disclosed herein can also be incorporated into the parenteral dosage forms provided herein.

Combination Treatment

Also contemplated herein are combination therapies, for example, co-administering a disclosed compound and an additional active agent, as part of a specific treatment regimen intended to provide the beneficial effect from the co-action of these therapeutic agents. The beneficial effect of the combination includes, but is not limited to, pharmacokinetic or pharmacodynamic co-action resulting from the combination of therapeutic agents. Administration of these therapeutic agents in combination typically is carried out over a defined time period (usually weeks, months, or years depending upon the combination selected). Combination therapy is intended to embrace administration of multiple therapeutic agents in a sequential manner, that is, wherein each therapeutic agent is administered at a different time, as well as administration of these therapeutic agents, or at least two of the therapeutic agents, in a substantially simultaneous manner.

Substantially simultaneous administration is accomplished, for example, by administering to the subject a single formulation or composition, (e.g., a tablet or capsule having a fixed ratio of each therapeutic agent or in multiple, single formulations (e.g., capsules) for each of the therapeutic agents. Sequential or substantially simultaneous administration of each therapeutic agent is effected by any appropriate route including, but not limited to, oral routes, intravenous routes, intramuscular routes, and direct absorption through mucous membrane tissues. The therapeutic agents are administered by the same route or by different routes. For example, a first therapeutic agent of the combination selected is administered by intravenous injection while the other therapeutic agents of the combination are administered orally. Alternatively, for example, all therapeutic agents are administered orally or all therapeutic agents are administered by intravenous injection.

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Combination therapy also embraces the administration of the therapeutic agents as described above in further combination with other biologically active ingredients and non-drug therapies. Where the combination therapy further comprises a non-drug treatment, the non-drug treatment is conducted at any suitable time so long as a beneficial effect from the co-action of the combination of the therapeutic agents and non-drug treatment is achieved. For example, in appropriate cases, the beneficial effect is still achieved when the non-drug treatment is temporally removed from the administration of the therapeutic agents, perhaps by days or even weeks.

The components of the combination are administered to a patient simultaneously or sequentially. It will be appreciated that the components are present in the same pharmaceutically acceptable carrier and, therefore, are administered simultaneously. Alternatively, the active ingredients are present in separate pharmaceutical carriers, such as conventional oral dosage forms, that are administered either simultaneously or sequentially.

For example, e.g., for contemplated treatment of pain, a disclosed compound is co-administered with another therapeutic for pain such as an opioid, a cannabinoid receptor (CB-1 or CB-2) modulator, a COX-2 inhibitor, acetaminophen, and/or a non-steroidal anti-inflammatory agent. Additional therapeutics e.g., for the treatment of pain that are co-administered, include morphine, codeine, hydromorphone, hydrocodone, oxycodone, fentanyl, tramadol, and levorphanol.

Other contemplated therapeutics for co-administration include aspirin, naproxen, ibuprofen, salsalate, diflunisal, dexibuprofen, fenoprofen, ketoprofen, oxaprozin, loxoprofen, indomethacin, tolmetin, sulindac, etodolac, ketorolac, piroxicam, meloxicam, tenoxicam, droxicam, lornoxicam, celecoxib, parecoxib, rimonabant, and/or etoricoxib.

EXAMPLES

Example 1: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A

Form A was obtained with several solvents, including methanol, isopropyl acetate, and a 9:1 mixture of ethanol/water. Form A was also prepared by slurrying starting material (a mixture of Form A/C) in toluene at RT for 1 day. NMR spectra of Form A indicates trace amount of surface toluene. HPLC purity of Form A sample was measured to be 99.87 area%. Form A has a differential scanning calorimetry thermogram comprising an endotherm with an onset of about 111-112° C. and a maximum at about 113° C. (shown in FIG. 2). Form A has a thermogravimetric analysis thermogram comprising a weight loss of less than about 0.1% when heated from about 30° C. to about 100° C., indicating anhydrous material (shown in FIG. 2).

The dynamic vapor sorption (DVS) isotherm plot of Form A was collected at 25° C. between 0 and 95% RH (shown in FIG. 3). A water uptake of 0.05% was observed at 25° C./80% RH. The XRPD comparison indicated that no form change was observed for Form A after the DVS test (shown in FIG. 4).

Form A sample showed good physical and chemical stability under accelerated conditions of 25° C./60% RH and 40° C./75% RH for 3 months. X-ray powder diffractograms comparing Form A before and after accelerated conditions is shown in FIG. 5. XRPD peaks for Form A are shown in Tables 1 and 2 (different XRPD instruments and samples; peaks below a certain amplitude are omitted for clarity).

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TABLE 1

XRPD peaks of Form A.			
No.	2-theta(deg)	d(ang.)	Height(counts)
1	4.9	17.85	445
2	9.9	8.906	7062
3	12.5	7.086	450
4	13.4	6.625	1881
5	14.0	6.324	3185
6	14.2	6.212	3613
7	14.9	5.934	4377
8	16.0	5.545	1117
9	16.5	5.378	3668
10	16.7	5.302	461
11	17.3	5.131	2862
12	18.1	4.888	10138
13	18.8	4.721	543
14	19.3	4.589	2757
15	19.7	4.50	561
16	19.9	4.449	6638
17	20.1	4.405	12550
18	20.6	4.311	3593
19	21.4	4.151	2679
20	22.3	3.975	21984
21	23.1	3.851	3116
22	24.1	3.684	1875
23	24.7	3.605	4728
24	25.1	3.551	636
25	26.1	3.415	947
26	27.0	3.301	2208
27	27.2	3.279	2457
28	28.0	3.185	852
29	28.8	3.100	535
30	29.3	3.041	1477
31	31.2	2.863	536
32	32.2	2.777	594
33	34.3	2.615	457
34	36.7	2.446	592
35	38.2	2.352	607

TABLE 2

XRPD peaks of Form A.			
No.	2-theta(deg)	d(ang.)	Height(counts)
1	4.9	17.85	445
2	9.9	8.906	7062
3	12.5	7.086	450
4	13.4	6.625	1881
5	14.0	6.324	3185
6	14.2	6.212	3613
7	14.9	5.934	4377
8	16.0	5.545	1117
9	16.5	5.378	3668
10	16.7	5.302	461
11	17.3	5.131	2862
12	18.1	4.888	10138
13	18.8	4.721	543
14	19.3	4.589	2757
15	19.7	4.50	561
16	19.9	4.449	6638
17	20.1	4.405	12550
18	20.6	4.311	3593
19	21.4	4.151	2679
20	22.3	3.975	21984
21	23.1	3.851	3116
22	24.1	3.684	1875
23	24.7	3.605	4728
24	25.1	3.551	636
25	26.1	3.415	947
26	27.0	3.301	2208
27	27.2	3.279	2457
28	28.0	3.185	852
29	28.8	3.100	535
30	29.3	3.041	1477

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TABLE 2-continued

XRPD peaks of Form A.			
No.	2-theta(deg)	d(ang.)	Height(counts)
31	31.2	2.863	536
32	32.2	2.777	594
33	34.3	2.615	457
34	36.7	2.446	592
35	38.2	2.352	607

Example 2: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form B

Form B was obtained with a 9:1 mixture of acetonitrile/water. Form B was also obtained via liquid vapor diffusion from mixture of ethyl acetate/n-heptane as well as evaporation in neat methyl acetate, acetone, or methylene chloride. The XRPD peaks for Form B are displayed in Table 3. A weight loss of less than 1.0% was observed when heated from about 30° C. to about 100° C. The DSC result exhibited a sharp endotherm onset at 91° C. and an endotherm with a maximum at about 95° C. Based on the low TGA weight loss and the single sharp DSC endotherm, Form B was postulated to be an anhydrate.

TABLE 3

XRPD peaks of Form B.			
No.	2-theta(deg)	d(ang.)	Height(counts)
1	6.9	12.86	341
2	9.1	9.74	232
3	10.4	8.522	563
4	11.5	7.700	142
5	12.7	6.985	359
6	13.1	6.730	257
7	13.8	6.39	273
8	14.3	6.182	473
9	15.1	5.847	2206
10	16.2	5.465	8503
11	17.3	5.114	551
12	17.9	4.950	2243
13	18.2	4.871	9600
14	18.7	4.754	4314
15	19.2	4.619	4926
16	20.1	4.421	3639
17	20.7	4.289	859
18	20.8	4.262	1649
19	21.1	4.205	2422
20	21.3	4.165	1181
21	21.6	4.120	421
22	22.7	3.922	1134
23	23.0	3.861	720
24	24.3	3.661	1424
25	25.0	3.566	2080
26	25.4	3.498	525
27	25.8	3.458	460
28	27.0	3.301	568
29	27.4	3.252	1417
30	27.6	3.224	827
31	28.4	3.143	238
32	29.4	3.036	402
33	29.9	2.991	128
34	30.6	2.924	715
35	31.7	2.823	129
36	33.5	2.673	139
37	34.8	2.575	198
38	37.0	2.431	179
39	39.5	2.282	190

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Example 3: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form C

Form C was obtained when Form B was slurried in FaSSIF-v2 or FeSSIF-v2 at 37° C. for 2 hr. Form C can also be obtained with a 1:3 chloroform/hexane mixture. The DSC shows an endotherm with a maximum at about 99° C., and the TGA shows a weight loss of less than about 0.15% when heated from about 30° C. to about 150° C., indicating Form C is an anhydrate. XRPD peaks for Form C are shown in Table 4.

TABLE 4

XRPD peaks of Form C.			
No.	2-theta(deg)	d(ang.)	Height(counts)
1	5.1	17.18	1235
2	8.5	10.42	146
3	10.2	8.633	661
4	10.4	8.463	402
5	10.6	8.340	697
6	10.8	8.179	426
7	12.3	7.209	4781
8	12.6	7.007	943
9	14.1	6.294	1700
10	14.7	6.021	1871
11	14.8	5.963	2174
12	15.4	5.747	691
13	15.5	5.699	2577
14	15.7	5.628	1504
15	16.2	5.452	2030
16	16.8	5.259	5422
17	17.1	5.194	1475
18	17.3	5.135	611
19	18.0	4.928	9698
20	18.2	4.876	4834
21	18.5	4.798	6212
22	18.7	4.752	576
23	19.5	4.551	6357
24	20.0	4.444	11680
25	20.5	4.333	2281
26	21.0	4.223	15490
27	21.4	4.144	6065
28	21.8	4.070	1064
29	22.2	3.994	1216
30	22.8	3.893	4382
31	23.3	3.808	2959
32	23.8	3.736	989
33	24.6	3.616	5533
34	25.5	3.490	6853
35	25.9	3.434	3080
36	26.4	3.375	372
37	26.6	3.348	605
38	27.0	3.305	1594
39	27.6	3.226	229
40	28.0	3.186	201
41	28.2	3.159	248
42	29.6	3.019	2177
43	31.2	2.868	498
44	31.3	2.852	398
45	32.0	2.797	477
46	32.8	2.728	299
47	33.3	2.692	205
48	34.2	2.621	165
49	34.5	2.598	269
50	36.0	2.490	417
51	36.5	2.462	472
52	37.2	2.418	92

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Example 4: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form D

Form D can be obtained from slurrying Form B in MTBE at room temperature for 10 days. The DSC shows an endotherm with a maximum at about 101° C., and the TGA shows a weight loss of about 0.054% when heated from about 30° C. to about 150° C., indicating Form D is an anhydrate. XRPD peaks for Form D are shown in Table 5.

TABLE 5

XRPD peaks of Form D.			
No.	2-theta(deg)	d(ang.)	Height (counts)
1	5.9	15.094	2047
2	8.4	10.530	900
3	8.8	10.088	630
4	9.5	9.323	538
5	11.7	7.558	498
6	13.4	6.625	8528
7	14.0	6.305	2653
8	14.3	6.183	716
9	14.8	5.985	2408
10	15.3	5.794	365
11	16.1	5.510	4141
12	16.8	5.284	2986
13	17.8	4.981	907
14	18.0	4.924	1322
15	18.3	4.831	6368
16	18.6	4.762	2087
17	18.8	4.707	634
18	19.5	4.553	2557
19	19.8	4.488	2428
20	20.7	4.291	4145
21	21.0	4.218	11450
22	22.0	4.033	1042
23	22.8	3.893	1266
24	23.0	3.859	1217
25	23.8	3.734	23683
26	24.4	3.643	3299
27	24.8	3.588	1917
28	25.5	3.491	514
29	26.3	3.389	1610
30	26.7	3.341	8142
31	26.9	3.313	1471
32	28.0	3.187	402
33	29.5	3.021	5758
34	32.5	2.755	965
35	33.3	2.688	774
36	34.1	2.630	168
37	34.5	2.597	409
38	34.8	2.577	418
39	35.0	2.560	253
40	36.2	2.480	555
41	36.9	2.433	224
42	37.2	2.413	319
43	37.8	2.376	248
44	38.6	2.329	294
45	39.6	2.276	359

Example 5: (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form E

Form E can be obtained from slurring Form C in a chloroform/hexane mixture at rt for 9 days. The DSC shows an endotherm with a maximum at about 104° C., and the TGA shows a weight loss of about 0.37% when heated from about 30° C. to about 150° C., indicating Form E is an anhydrate. XRPD peaks for Form E are shown in Table 6.

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TABLE 6

XRPD peaks for Form E.			
No.	2-theta(deg)	d(ang.)	Height (counts)
1	7.5	11.73	333
2	9.1	9.71	115
3	10.9	8.144	698
4	12.7	7.006	1164
5	14.0	6.327	793
6	14.3	6.173	1113
7	15.8	5.622	2425
8	16.2	5.462	915
9	16.8	5.279	274
10	17.2	5.158	1845
11	17.8	4.985	2556
12	18.4	4.818	1879
13	19.7	4.502	1215
14	20.3	4.371	778
15	21.3	4.167	2749
16	21.9	4.054	2304
17	22.7	3.923	377
18	22.8	3.892	5540
19	23.5	3.790	358
20	24.2	3.681	1614
21	25.4	3.511	392
22	25.6	3.475	839
23	26.2	3.399	398
24	27.3	3.264	171
25	28.3	3.156	237
26	29.0	3.075	342
27	29.7	3.007	278
28	30.0	2.975	310
29	30.7	2.910	151

Example 6: Stability and Solubility Tests

A solution stability test of Form A was performed to assess substantial degradation of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate in physiologically relevant media. A solution of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate (1 mg/mL) was prepared in acetonitrile/water (50/50 v/v). Test medium (1990 μ L) was added to a vial. Test media evaluated included buffers pH 1-8, simulated gastric fluid (SGF), fasted state simulated intestinal fluid (FaSSIF), and fed state simulated intestinal fluid (FeSSIF). Stock solution (10 μ L) was added to the vial containing the medium. This mixture was stirred to mix well and transferred to a HPLC vial, which was placed in the autosampler maintained at 37° C. The vial was injected repeatedly every 4 hr for 24 hr. Results are shown in Table 7. Solubility is shown in Table 8.

TABLE 7

Solution chemical stability data for crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl- 4-(3-(trifluoromethoxy)benzyl) piperazine-1-carboxylate Form A at 37° C.								
Time (hr)	pH = 1	pH = 3	pH = 5.0	pH = 6.8	pH = 8.0	SGF	FaSSIF	FeSSIF
0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
4	99.6	99.8	100.9	100.2	101.3	100.2	100.6	99.6
8	100.3	99.8	101.7	100.8	100.5	100.1	100.2	100.2
12	100.3	98.5	99.9	100.1	99.7	99.2	100.5	99.7
16	99.9	99.3	100.7	100.0	100.1	99.7	100.3	99.6
20	101.9	99.6	99.9	99.6	98.7	100.4	100.0	99.8
24	100.5	99.5	99.6	99.0	99.4	99.6	100.0	99.7

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TABLE 8

Solubility of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A in physiologically relevant media (37° C.).		
Medium	Final Measured pH	Solubility (mg/mL)*
pH 1.0 HCl/NaCl buffer	1.47	>20
pH 2.5 Phosphate buffer	3.44	4.326
pH 3 Phosphate buffer	4.20	1.251
pH 4 Acetate buffer	4.18	0.696
pH 5 Acetate buffer	5.14	0.091
pH 6.8 Phosphate buffer	6.88	0.007
pH 8.0 Phosphate buffer	8.10	0.005
SGF with pepsin	1.75	>20
FaSSIF	6.48	0.146
FeSSIF	5.17	1.018
pH 1 (0.1N HCl)	1.48	>20
pH 2 (0.01N HCl)	3.55	3.858
pH 3 (0.001N HCl)	4.90	0.057
pH 4 (0.0001N HCl)	7.78	0.011
Water	8.10	0.012

Example 7: Stress Polymorph Screen

A stress polymorph screen using Form B as the starting form was performed. The tests were performed at room temperature or lower in a variety of solvents. Results are shown in Table 9.

TABLE 9

Stress Polymorph Screen Using Form B as the Starting Form at room temperature or lower.				
Experiment	Solvent	Temp.	State	Result
1	Ethyl Acetate	Room*	Slurry	Form A
2	Isopropyl Acetate	Room	Slurry	Form A
3	Acetone	4° C.	Evaporation	Form B
4	Dichloromethane	4° C.	Evaporation	Form B
5	Acetonitrile	Room*	Slurry	Form A
6	2-Me THF	4° C.	Evaporation	Form A
7	H ₂ O/MeOH 1/3 (v/v) Aw~0.5	Room*	Slurry	Form A
8	H ₂ O/MeCN 1/3 (v/v) Aw~0.9	4° C.	Evaporation	Mainly Form C
9	MTBE	Room	Slurry	Form D
10	EtOH	Room*	Slurry	Form A
11	2-Propanol	Room*	Slurry	Form A
12	Nitromethane	Room*	Slurry	Form A
13	H ₂ O	Room	Slurry	Form A

TABLE 9-continued

Stress Polymorph Screen Using Form B as the Starting Form at room temperature or lower.				
Experiment	Solvent	Temp.	State	Result
14	DMSO drops dispersed into water	Room	Slurry	Form B

*Clear solution observed initially; continued stirring @ 4° C. for 1 day followed by RT for 9 days

Example 8: Polymorph Screen

Polymorph screening at 50° C. was performed. A solution starting with Form B was prepared in a concentration of about 50 mg in 100 to 300 µL of solvents listed in Table 10. The solution was stirred at 50° C. for 24 h.

TABLE 10

Polymorph Screening at 50° C.		
Condition	State	Result
H ₂ O	Slurry	Form A
MTBE	Slurry	Form A
CPME	Clear solution	Evaporation clear film only

Example 9: Lyophilization Study

Attempts were made to generate amorphous material by lyophilizing Form B. Form B was lyophilized with a LyoSys 3 freeze dryer, in DPD core area at -40° C. overnight. Results are shown Table 11.

TABLE 11

Lyophilization attempts.		
Experiment	Form/Solvent	Result
1	240 mg Form B in~4:1 H ₂ O:MeCN	Form A
2	240 mg Form B in~3:2 H ₂ O:MeCN	Form B
3	240 mg Form B in~2:1 H ₂ O:MeCN	Form B

Example 10: Form A Solubility Study

Solubility was evaluated at 2 hr and 24 hr for Form A in FaSSIF-v2 and FeSSIF-v2 at 37° C. Approximately 40 mg of Form A sample was added in 2.0 mL of corresponding bio-medium to form a suspension, followed by shaking at

500 rpm at 37° C. After 2 hr and 24 hr, the supernatants were isolated by centrifugation (7000 rpm for 5 min) followed by filtration through 0.45 mm PTFE membrane and were tested by HPLC concentration and pH. The residual solids were characterized by XRPD. As summarized in Table 12, the 24

no significant form change or HPLC purity decrease was observed after 3 months at both conditions. Therefore, Form A sample showed good physical and chemical stability under 25° C./60% RH and 40° C./75% RH conditions for 3 months.

TABLE 13

Stability results summary.						
Time point	Condition	Crystal form	TGA weight loss (% up to 100° C.)	DSC endotherm	HPLC Purity (area %)	Assay vs. 5° C. control (%)
Initial	—	Form A	1.2	110.9	99.86	NA
2 weeks	25° C./60% RH	Form A	1.1	110.1	99.84	101.2
	40° C./75% RH	Form A	1.1	110.5	99.78	98.1
4 weeks	25° C./60% RH	Form A	1.7	110.9	99.86	101.4
	40° C./75% RH	Form A	1.9	110.8	99.86	101.3
6 weeks	25° C./60% RH	Form A	0.2	110.4	99.86	99.7
	40° C./75% RH	Form A	0.8	110.8	99.86	99.2

Time point	Condition	Crystal form	TGA weight loss (% up to 100° C.)	DSC endotherm	HPLC Purity (area %)	Assay vs. 5° C. control (%)	Chiral Purity
Initial	NA	Form A	1.2	110.9	99.86	NA	100*
3 months	25° C./60% RH	Form A	0.03	112.6	99.86	97.3	100
	40° C./75% RH	Form A	0.01	112.9	99.86	96.9	100

*The chiral purity was tested for the starting material after stored at -20° C. for about 3 months

hr solubility of Form A in FaSSiF-v2 and FeSSiF-v2 was measured to be 0.063 and 1.1 mg/mL, respectively. The XRPD patterns indicated that no form change was observed after solubility evaluation.

TABLE 12

Summary of solubility evaluation for Form A at 37 ° C.					
Medium	Time point (hr)	Solubility (mg/mL)	Observation	Final Form	Final pH
FaSSiF (pH 6.5)	2	0.063	Turbid	Form A	6.5
FaSSiF (pH 6.5)	24	0.063	Turbid	Form A	6.6
FeSSiF (pH 5.8)	2	1.0	Turbid	Form A	5.9
FeSSiF (pH 5.8)	24	1.1	Turbid	Form A	6.0

Example 11: Accelerated Solid-State Stability Study of Form A

Solid-state stability evaluation (accelerated conditions) was performed on Form A under conditions of 25° C./60% RH and 40° C./75% RH. The detailed procedure utilized for stability evaluation is listed below:

- 1) weigh approximately 30 mg of Form A sample into each 3-mL vial;
- 2) cover the vial with Parafilm® and poke 15 holes on the Parafilm® with a needle;
- 3) put the vial in a stability chamber of 25° C./60% RH or 40° C./75% RH; and
- 4) sample for XRPD, TGA, DSC, HPLC purity and assay tests after 2, 4 and 6 weeks and 3 months.

The results of solid-state stability evaluation are summarized in Table 13. The impurities are summarized in Table 14. Based on (RFD, TGA, and DSC characterization results,

TABLE 14

% Area of impurities summary in solid-state stability study.					
Time point	Condition	% Area			
		Imp.1 (RRT 0.81)	API (RRT 1.00)	Imp.2 (RRT 1.04)	Imp.3 (RRT 1.74)
Initial	NA	0.09	99.86	—	0.05
2 weeks	25° C./60% RH	0.07	99.84	—	0.09
	40° C./75% RH	0.08	99.78	0.07	0.07
4 weeks	25° C./60% RH	0.07	99.86	—	0.07
	40° C./75% RH	0.07	99.86	—	0.07
6 weeks	25° C./60% RH	0.07	99.86	—	0.07
	40° C./75% RH	0.07	99.86	—	0.07

Time point	Condition	% Area		
		Imp.1 (RRT 0.81)	API (RRT 1.00)	Imp.3 (RRT 1.74)
Initial	NA	0.09	99.86	0.05
3 months	25° C./60% RH	0.09	99.86	0.05
	40° C./75% RH	0.09	99.86	0.05

—: <0.05.

Example 12: Solid State Characterization of Form A

A summary of baseline solid state characterization results of crystalline (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate Form A, including pKa, partition coefficient (log D7.4) measurement and 2-hrs equilibrium solubility in FaSSiF and FeSSiF (version 2) are provided herein. All results obtained are summarized in Table 15. The pKa values were first predicted by Marvin® and then detected by Sirius T3 with the range from 2.0 to 12.0 (in triplicate). Log D7.4 was measured with the solvent system of pH 7.4 phosphate buffer and n-octanol by shake-flask method. Meanwhile, 2-hrs equilibrium solubility of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trif-

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luoromethoxy)benzyl)piperazine-1-carboxylate in FaSSIF and FeSSIF (version 2) was measured to be 0.12 and 0.080 mg/mL.

TABLE 15

Summary of pKa, Log D and solubility results.					
pKa			2-Hrs equilibrium		
Predicted	Measured		solubility (mg/mL)		
pKa value	pKa value	Log D _{7.4}	FaSSIF	FeSSIF	
2.26, 6.67	5.79	3.76	0.12	0.080	

pKa values were predicted and detected in the range of 2.0 to 12.0

N/A: no measurement.

Example 13: Solubility of Form A

Approximate solubility of Form A was determined in 20 solvent systems at RT. Approximately 2 mg of sample was added into a 3-mL glass vial. Solvents in Table 16 were then added stepwise (100 μ L per step) into the vials until the solids were dissolved visually or a total volume of 2 mL was reached. Solubility results summarized in Table 16 were used to guide the solvent selection in screening experiment design.

TABLE 16

Approximate solubility of Form A at RT			
Solvent	Solubility (mg/mL)	Solvent	Solubility (mg/mL)
MeOH	S > 23.0	1,4-dioxane	S > 22.0
EtOH	S > 17.0	DCM	S > 19.0
IPA	S > 24.0	CHCl ₃	S > 19.0
Acetone	S > 17.0	ACN	S > 24.0
MIBK	S > 24.0	Anisole	S > 20.0
EtOAc	S > 26.0	toluene	4.5 < S < 6.0
IPAC	S > 21.0	n-heptane	S < 0.9
MTBE	7.7 < S < 11.5	NMP	S > 21.0
THF	S > 26.0	DMSO	S > 20.0
2-MeTHF	S > 23.0	H ₂ O	S > 1.0

Example 14: Solubility of Form B

Solubility of Form B was evaluated at 2 hr for Form B in FaSSIF-v2 and FeSSIF-v2 at 37° C. Approximately 10 to 20 mg of Form B sample was added in 0.8 mL of corresponding bio-relevant medium to form a suspension, followed by shaking at 500 rpm at 37° C. After 2 hr, the supernatants were isolated by centrifugation (7200 rpm for 5 min) followed by filtration through 0.45 mm PTFE membrane and were tested by HPLC concentration and pH. The residual solids were characterized by XRPD. As summarized in Table 17, Form B converted to a new form (Form C) after solubility evaluation, while the solubility in FaSSIF-v2 and FeSSIF-v2 was measured to be 0.12 and 0.080 mg/mL, respectively.

TABLE 17

Summary of solubility evaluation for Form B at 37° C.					
Medium	Time point (hr)	Solubility (mg/mL)	Observation	Final Form	Final pH
FaSSIF (pH 6.5)	2	0.12	Turbid	Form C	6.6

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TABLE 17-continued

Summary of solubility evaluation for Form B at 37° C.					
Medium	Time point (hr)	Solubility (mg/mL)	Observation	Final Form	Final pH
FeSSIF (pH 5.8)	2	0.08	Turbid	Form C	5.9

Approximately 55 to 90 mg of Form B was added into a 4 mL glass vial. Solvent list in Table 18 was added into each of the vial. These samples were stirred for 3 minutes prior to visual solubility assessment.

TABLE 18

Summary of solubility evaluation for Form B at room temperature.		
1	Ethyl Acetate	>800 mg/mL
2	Isopropyl Acetate	<810 mg/mL
3	Acetone	>430 mg/mL
4	Dichloromethane	>830 mg/mL
5	Acetonitrile	>720 mg/mL
6	2-Me THF	>810 mg/mL
7	H ₂ O/MeOH 1/3 (v/v) Aw~0.5	>780 mg/mL
8	H ₂ O/MeCN 1/3 (v/v) Aw~0.9	>730 mg/mL
10	EtOH	>800 mg/mL
11	2-Propanol	>365 mg/mL
12	Nitromethane	>560 mg/mL

Example 15: Anti-solvent Addition Experiments

Anti-solvent addition experiments were performed. A total of 8 anti-solvent addition experiments were carried out. Approximately 15 mg of Form A was dissolved in 0.5 mL solvent to obtain a clear solution, and the solution was magnetically stirred, followed by the addition of the anti-solvent to induce precipitation until the total amount of anti-solvent reached 10 mL. The precipitate was isolated for XRPD analysis. Clear solutions were transferred to -20° C. for 5 hr. Results summarized in Table 19 showed that only Form A was obtained.

TABLE 19

Summary of anti-solvent addition experiments.		
Solvent	Anti-solvent	Solid Form
EtOH	H ₂ O	Form A*
1,4-dioxane		Form A
DMSO		Form A
THF		Form A
EtOAc		Form A
CHCl ₃	n-heptane	Form A*
Acetone		Form A*
THF		Form A*

*: These solids were obtained by slurry at -20° C.

Example 16: Slow Evaporation Experiments

Slow evaporation experiments were performed under 6 conditions. Approximately 15 mg of Form A was dissolved in 1.0 mL of corresponding solvent in a 3-mL glass vial. The visually clear solutions were evaporated at RT to induce precipitation. The solids were isolated for XRPD analysis. Results summarized in Table 20 indicated that Form A, Form B, and amorphous were obtained.

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TABLE 20

Summary of slow evaporation experiments.	
Solvent (v/v)	Solid Form
MeOH	Form A
IPAC	Form A
2-MeTHF	Amorphous
ACN	Amorphous
EtOH:H ₂ O (9:1)	Form A
ACN:H ₂ O (9:1)	Form B

Example 17: Slow Cooling Experiments

Slow cooling experiments were conducted in 2 solvent systems. Approximately 50 mg of Form A was suspended in 0.51.0 mL of solvent in a 3-mL glass vial at RT. The suspension was then heated to 50° C., equilibrated for about 2 hr and filtered into a new vial. Filtrates were slowly cooled down to 5° C. at a rate of 0.1° C./min. Clear solutions were transferred to -20° C., followed by evaporation to dryness at RT and then solids were tested by XRPD. Results summarized in Table 21 indicated only Form A was obtained.

TABLE 21

Summary of slow cooling experiments.	
Solvent (v/v)	Solid Form
MTBE/EtOAc (9:1)	Form A*
Toluene/EtOH (9:1)	Form A*

*: These solids were obtained by evaporation at RT

Example 18: Slurry Conversion Experiments

Slurry conversion experiments were performed. Slurry conversion experiments were conducted at RT in 7 different solvent systems. Approximately 20 mg of Form A was suspended in 0.30.9 mL of solvent in a HPLC glass vial. After the suspensions were stirred for 5 days at RT, the remaining solids were isolated for XRPD analysis. Clear solutions were obtained after stirring, in which anti-solvent of n-heptane was added to induce crystallization. Results summarized in Table 22 indicated that only Form A was obtained.

TABLE 22

Summary of slurry conversion experiments at RT.	
Solvent (v/v)	Solid Form
MTBE	Form A
Toluene	Form A
Acetone/n-heptane (1:2)	Form A
THF/n-heptane (1:2)	Form A*
1,4-dioxane/H ₂ O (1:2)	Form A
IPA/H ₂ O (937/63, aw~0.5)	Form A
H ₂ O	Form A

*: Solids were obtained after anti-solvent addition of n-heptane.

Slurry conversion experiments were also conducted at 50° C. in 3 different solvent systems. Approximately 20 mg of Form A was suspended in 0.41.0 mL of solvent in a HPLC glass vial. After the suspensions were stirred for 5 days at 50° C., the remaining solids were isolated for XRPD analysis. Clear solutions were obtained after stirring, in which

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anti-solvent of n-heptane was added to induce crystallization. Results summarized in Table 23 showed only Form A was obtained.

TABLE 23

Summary of slurry conversion experiments at 50° C.	
Solvent (v/v)	Solid Form
MTBE	Form A*
Toluene	Form A*
H ₂ O	Form A

*: Solids were obtained after anti-solvent addition of n-heptane.

Example 19: Liquid Vapor Diffusion Experiments

Liquid Vapor Diffusion experiments were performed. Four liquid vapor diffusion experiments were conducted. Approximately 15 mg of Form A was dissolved in 0.5 mL of solvent to obtain a clear solution in a 3-mL vial. This solution was then placed into a 20-mL vial with 4 mL of anti-solvent. The 20-mL vial was sealed with a cap and kept at RT for organic vapor to interact with the solution. The precipitates were isolated for XRPD analysis. The results summarized in Table 24 showed that Form A and Form B were obtained.

TABLE 24

Summary of liquid vapor diffusion experiments.		
Solvent	Anti-solvent	Solid Form
THF	H ₂ O	Form A
EtOH		Form A
EtOAc		Form B
CHCl ₃	n-heptane	Form A

Example 20: Interconversion Study

An interconversion study of crystal forms was performed. To understand the thermodynamic stability relationship between Form A and B, competitive slurry experiments were performed at RT and 50° C. in water and toluene. Prior to the study, Form A was used to saturate the corresponding solvent before it was filtered to obtain a near-saturated solution. Equal amounts of Form A and B samples were weighed and then mixed with the prepared near-saturated solution to form a new suspension, which was stirred magnetically (~1000 rpm) at RT and 50° C. As summarized in Table 25, Form A was obtained after slurry at RT and 50° C. for 1-3 weeks, indicating that Form A was thermodynamically more stable than Form B from RT to 50° C.

TABLE 25

Thermodynamic stability study.			
Starting Form	Solvent (v/v)	Temperature	Final Form
Form A + B	H ₂ O/PhMe	RT	Form A
Form A + B	H ₂ O/PhMe	50° C.	Form A

Example 21: Competitive Slurry Study

A competitive slurry study of all known forms A to E was performed in water at room temperature to understand the

thermodynamic stability relationship among these forms. Form A was used to saturate the water before it was filtered to obtain a near-saturated solution. Equal amounts of each form (A to E) were weighed and then mixed with the prepared near-saturated solution to form a new suspension, which was stirred magnetically (~1000 rpm) at room temperature for 7 days. Solids were isolated and analyzed by XRPD and Form A was obtained indicating that Form A was thermodynamically the most stable form at RT.

Example 22: Vapor Induced Crystallization Experiments

A total of 9 different experiments were conducted. About 100 mg of Form B was placed in a 4 mL glass vial which was then placed into a 20-mL vial with 4 mL of anti-solvent in it. The 20-mL vial was sealed with a cap and kept at RT for organic vapor to interact with the solid. After 7 to 10 days, wet solids were isolated and analyzed by XRPD. The results summarized in Table 26 showed that Form A and Form B were obtained.

TABLE 26

Summary of vapor induced crystallization experiments	
Condition	Result
Ethanol	Form A
DMSO	Form A
Isopropyl Acetate	Form A
Methyl Acetate	Form B
THF	Clear film after evaporation
CH ₂ Cl ₂	
Cyclohexanone	
Dimethyl Formamide	
N-butyl Acetate	

Characterization Methodology

Samples generated as described in the solid form screen were typically analyzed by X-Ray Powder Diffraction (XRPD). XRPD was conducted on a PANalytical Emptyrean and X'Pert3 X-ray powder diffractometer using Cu K α radiation at 1.54 Å. In general, positions of XRPD peaks are expected to individually vary on a measurement-by-measurement basis by about $\pm 0.2^\circ 2\theta$. In general, as understood in the art, two XRPD patterns match one another if the characteristic peaks of the first pattern are located at approximately the same positions as the characteristic peaks of the second pattern. As understood in the art, determining whether two XRPD patterns match or whether individual peaks in two XRPD patterns match may require consideration of individual variables and parameters such as, but not limited to, preferred orientation, phase impurities, degree of crystallinity, particle size, variation in diffractometer instrument setup, variation in XRPD data collection parameters, and/or variation in XRPD data processing, among others. The determination of whether two patterns match may be performed by eye and/or by computer analysis. An example of an XRPD pattern collected and analyzed using these methods and parameters is provided herein, e.g., as FIG. 1.

PANalytical Emptyrean

Scan Mode: Continuous

Detector: D/teX Ultra

Divergence slit: Automatic

Start Angle: 3 deg.

Stop Angle: 40 deg.

Step Width: 0.0200 deg.

Scan time (s): 5.00 deg./min

Incident Slit: $\frac{1}{2}$ deg.

Receiving Slit #1: 8.000 mm

Receiving Slit #2: 13.000 mm

Wavelength: K α 1: 1.540598 Å; K α 2: 1.5444426 Å; inten-

5 sity ratio K α 2/K α 1: 0.50; 40 kV, 44 mA.

X'Pert3

Scan Mode: Continuous

Divergence slit: $\frac{1}{8}^\circ$

Start Angle: 3 deg

10 Stop Angle: 40 deg.

Step Size: 0.0263 deg.

Scan step time (s): 46.665

Test time: about 5 minutes

15 Wavelength: K α 1 : 1.540598 Å; K α 2: 1.5444426 Å; inten-

sity ratio K α 2/K α 1: 0.50

Differential Scanning calorimetry (DSC) analyses were performed on a TA Instruments Q200/Q2000. About 5 mg of sample was placed into a tared DSC crimped/open aluminum pan and the weight of the sample was accurately recorded. The heating rate was 10°C./min , using nitrogen as the purge gas. An example of a DSC thermogram collected and analyzed using these methods and parameters is provided herein, e.g., as FIG. 2.

Thermal Gravimetric Analyses (TGA) were performed on a TA Instruments Q500/Q5000. About 10 mg of sample was placed on an open aluminum pan, accurately weighed and loaded into the TGA furnace. The heating rate was 10°C./min , using nitrogen as the purge gas. An example of a TGA thermogram collected and analyzed using these methods and parameters is provided herein, e.g., as FIG. 2.

Dynamic vapor sorption (DVS) was performed on a Surface Measurement Systems DVS Intrinsic. The relative humidity at 25°C. were calibrated against deliquescence point of LiCl, Mg(NO₃)₂, and KCl. Sample sizes were between 10-20 mg. Gas and flow rate was nitrogen, 200 mL/min; dm/dt: 0.002%/min.

HPLC was performed with Agilent 1100/1260. The analysis for Form A was determined with a Phenomenex Gemini C18, 150 $\times$ 4.6 mm, 3 nm (column). The mobile phase was 0.1% formic acid in water; flow rate of 1.0 mL/min; column temperature of 40°C. ; and UV detector at 270 nm. The analysis for Form B was determined with a Kinetex 5 μm EVO C18, 150 $\times$ 4.6 mm, 5 μm (column). The mobile phase was 0.1% formic acid in acetonitrile; flow rate of 1.0 mL/min; column temperature of 40°C. ; and UV detector at 210 nm.

Proton NMR was collected on Bruker 400 NMR Spectrometer using DMSO-*d*₆.

The embodiments described above are intended to be merely exemplary, and those skilled in the art will recognize, or will be able to ascertain using no more than routine experimentation, numerous equivalents of specific compounds, materials, and procedures. All such equivalents are considered to be within the scope of the disclosure and are encompassed by the appended claims.

Citation or identification of any reference in this application is not an admission that such reference is available as prior art. The full scope of the disclosure is better understood with reference to the appended claims.

60 The invention claimed is:

1. A single unit dosage form comprising a crystalline form of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, wherein:
65 the crystalline form is Form A of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction

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- pattern comprising peaks at 18.1, 20.1, and 22.3 degrees $2\theta \pm 0.2$ degrees 2θ ;
 the crystalline form is Form B of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 16.2, 18.2, and 19.2 degrees $2\theta \pm 0.2$ degrees 2θ ;
 the crystalline form is Form C of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 18.0, 20.0, and 21.0 degrees $2\theta \pm 0.2$ degrees 2θ ;
 the crystalline form is Form D of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 13.4, 21.0, and 23.8 degrees $2\theta \pm 0.2$ degrees 2θ ; or
 the crystalline form is Form E of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 17.8, 21.3, and 22.8 degrees $2\theta \pm 0.2$ degrees 2θ .
2. The single unit dosage form of claim 1, wherein the crystalline form is Form A of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 18.1, 20.1, and 22.3 degrees $2\theta \pm 0.2$ degrees 2θ .
 3. The single unit dosage form of claim 2, wherein the X-ray powder diffraction pattern further comprises peaks at 9.9, 19.9, and 24.7 degrees $2\theta \pm 0.2$ degrees 2θ .
 4. The single unit dosage form of claim 1, wherein the crystalline form is Form B of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 16.2, 18.2, and 19.2 degrees $2\theta \pm 0.2$ degrees 2θ .
 5. The single unit dosage form of claim 4, wherein the X-ray powder diffraction pattern further comprises peaks at 15.1, 18.7, and 20.1 degrees $2\theta \pm 0.2$ degrees 2θ .
 6. The single unit dosage form of claim 1, wherein the crystalline form is Form C of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-car-

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- boxylate, having an X-ray powder diffraction pattern comprising peaks at 18.0, 20.0, and 21.0 degrees $2\theta \pm 0.2$ degrees 2θ .
7. The single unit dosage form of claim 6, wherein the X-ray powder diffraction pattern further comprises peaks at 12.3, 16.8, and 25.5 degrees $2\theta \pm 0.2$ degrees 2θ .
 8. The single unit dosage form of claim 1, wherein the crystalline form is Form D of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 13.4, 21.0, and 23.8 degrees $2\theta \pm 0.2$ degrees 2θ .
 9. The single unit dosage form of claim 8, wherein the X-ray powder diffraction pattern further comprises peaks at 18.3, 26.7, and 29.5 degrees $2\theta \pm 0.2$ degrees 2θ .
 10. The single unit dosage form of claim 1, wherein the crystalline form is Form E of (R)-5-carbamoylpyridin-3-yl-2-methyl-4-(3-(trifluoromethoxy)benzyl)piperazine-1-carboxylate, having an X-ray powder diffraction pattern comprising peaks at 17.8, 21.3, and 22.8 degrees $2\theta \pm 0.2$ degrees 2θ .
 11. The single unit dosage form of claim 10, wherein the X-ray powder diffraction pattern further comprises peaks at 15.8, 18.4, and 21.9 degrees $2\theta \pm 0.2$ degrees 2θ .
 12. The single unit dosage form of claim 1, wherein the single unit dosage form is suitable for oral, mucosal, rectal, nasal, vaginal, parenteral, subcutaneous, intramuscular, bolus injection, intraarterial, intravenous, sublingual, transdermal, buccal, or topical administration.
 13. The single unit dosage form of claim 12, wherein the single unit dosage form is a tablet, caplet, capsule, powder, or sterile solid that can be reconstituted to provide a liquid dosage form.
 14. The single unit dosage form of claim 12, wherein the single unit dosage form is a tablet, caplet, capsule, or liquid suitable for oral administration.
 15. The single unit dosage form of claim 12, wherein the single unit dosage form is a solution ready for injection, dry product ready to be dissolved or suspended in a pharmaceutically acceptable vehicle for injection, suspension ready for injection, or an emulsion ready for injection.
 16. The single unit dosage form of claim 1 comprising about 0.1 mg to about 1,000 mg of the crystalline form.

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